Images as Covariates: When do they Improve Prediction from Structured Data?

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Abstract

Images are increasingly used in economic research, but image-only prediction does not show whether images add information beyond existing covariates. We distinguish three questions: how well images predict outcomes, how much of that signal overlaps with structured data, and whether combining images with structured covariates improves out-of-sample prediction. A Monte Carlo examines whether and how to combine images with covariates; housing, horse performance, and academic salaries illustrate different cases. Images can add predictive value, provide redundant information, or help because they measure economically relevant attributes imperfectly captured in conventional data. Their value is therefore conditional on the analyst's information set.

Keywords: image data; unstructured data; generated covariates; prediction; hedonic models; transfer learning

JEL codes: C45, C53, C55, C81

1 Introduction

Researchers increasingly use photographs to supplement economic data. A common empirical mistake is to show that an image predicts an outcome and infer that adding images to an existing dataset must be useful. That inference does not follow. A photograph can predict salary because it reveals career stage or employer, both of which may already be recorded. It can predict house prices because it reveals size or location already captured by an MLS record. Image-only fit therefore cannot answer whether the photograph improves an existing empirical dataset.

The relevant comparison is between estimated models using structured variables X, image information I, and both $X+I$. Even a positive $X+I$ gain has more than one possible source. The structured information set may omit information relevant for predicting the outcome, or the estimated model for X may have the wrong functional form and miss information already contained in X. We address this distinction by measuring the image's contribution relative to a model using X. We choose this benchmark from linear, small neural-network (NN), and random-forest models using validation data. A random forest need not achieve the best possible prediction from X, but it lets us assess whether images still help when the structured model captures substantial nonlinearities and interactions.

We study three settings with different relationships between images and conventional data. For Toronto houses, exterior photographs supplement detailed property and neighbourhood characteristics in predicting sale prices. For Thoroughbred horses, standardized conformation photographs, which show body shape and proportions, supplement pedigree and auction-catalogue information in predicting workout performance. For Ontario university researchers, profile photographs are combined with citations, career-history information, and institution to predict salaries. Together, the applications range from settings where images may contain information missing from the structured data to one where the image is predictive on its own but largely redundant once conventional covariates are included.

Our Ontario salary application provides the central counterexample. Profile photographs of 2,320 university researchers predict log salary on their own, with a mean R^2 of 0.102 across repeated sample splits. Yet a model using citations, citation-history length, and institution fixed effects achieves an R^2 of 0.420. Adding image information using a combination method chosen on validation data lowers mean R^2 slightly to 0.418. Across 12 sample splits that keep repeated observations of the same researcher and duplicate images together, the increment ranges from -0.036 to 0.018. The photograph predicts salary on its own but does not improve

average out-of-sample performance when added to the estimated model using the existing covariates.

The results for house prices and horse workout speed show why this conclusion cannot be generalized. Two detached houses can have the same recorded bedrooms, bathrooms, lot frontage, sale month, and neighbourhood and still differ in maintenance, landscaping, additions, and privacy. An auction catalogue records a Thoroughbred's pedigree, sex, sale, and workout context but not every visually observable aspect. Images may recover outcome-relevant information not captured by these conventional variables. Whether they do is a question about modeling decisions as much as information sets, which we separate in Section 2.

We measure that incremental value by

$$\Delta R_I^2 = R_{X+I}^2 - R_X^2, \tag{1}$$

where both terms are evaluated on the same test observations, whose outcomes are not used to estimate or select the models. This comparison is specific to the estimated prediction methods. A positive value means that the estimated model using $X + I$ predicts better than the estimated model using X. It does not establish that the image provides information absent from X, identify a causal effect, or give the coefficient on an image-based prediction a structural interpretation.

In each application, we use 12 prespecified sample splits, allocating 70 percent of observations to training, 15 percent to validation, and 15 percent to testing. Repeated observations of the same unit and duplicate images remain in the same subsample. Within each split, we use the validation sample to choose the best model for the structured data and the best way to combine the structured and image information. The test partition remains untouched. Mean R^2 for X, I, and $X + I$ is 0.890, 0.677, and 0.897 for Toronto houses; 0.084, 0.044, and 0.108 for Thoroughbred speed; and 0.420, 0.102, and 0.418 for academic salaries. The mean increments are therefore 0.007, 0.024, and -0.002. Houses improve in 9 of 12 splits, horses in 11, and salaries in 8, although the salary changes remain centered near zero and include sizeable losses. These distributions show how results vary across sample splits; they are not confidence intervals.

The paper makes three contributions. First, it shows why image-only predictive success can mislead. Profile photographs alone attain a mean R^2 of 0.102 for Ontario academic salaries. Yet adding them to a flexible model using citations, citation-history length and

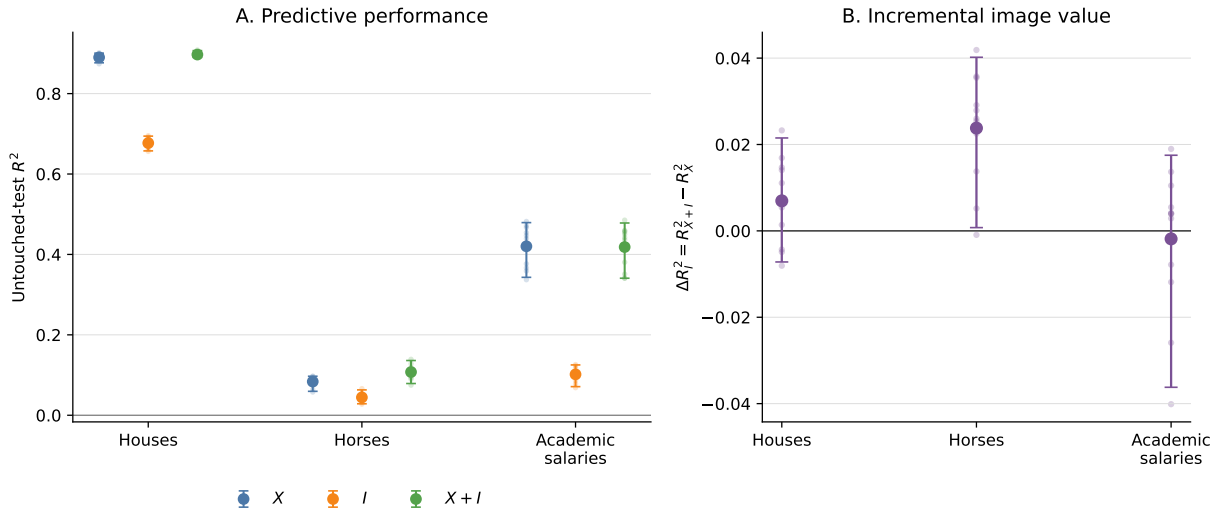


Figure 1: **Structured, image, and combined prediction across repeated splits.** Points in Panel A show mean test R^2 for the X, image, and $X + I$ models selected using validation data. Panel B reports ΔR_I^2. Small points show results from 12 prespecified splits that keep repeated units and duplicate images together; whiskers span the empirical 2.5th–97.5th percentiles. These ranges describe variation across sample splits and are not confidence intervals. Test outcomes select neither models nor splits.

institution lowers mean R^2 by 0.002. Second, it shows that an image's measured value depends on the benchmark it is measured against. On the fixed sample split for houses, the image increment falls from 0.032 against a linear structured model to 0.027 against a neural network and 0.013 against a random forest. This shows that the photograph's measured contribution can depend substantially on the flexibility of the structured model. Third, it shows that the way image and structured information are combined can change the answer. In the controlled experiment, a random forest using the structured data and a single image-based prediction attains a mean test R^2 of 0.610, compared with 0.485 for a neural model using the structured covariates and reduced image features. In the applications, validation favors models using structured covariates and reduced image features together for houses, but simpler models that combine separate predictions for horses. The contribution is not a new prediction estimator, and it is not the observation that conditioning matters. It is a comparison of settings in which researchers may consider adding images to an existing structured dataset.

This question builds on work using satellite, street-view, and property images to measure economic conditions and the built environment (Donaldson and Storeygard, 2016; Jean et al., 2016; Yeh et al., 2020; Glaeser et al., 2018; Dubey et al., 2016; Law et al., 2019). Modern

property applications combine visual and conventional information for valuation (Yazdani and Raissi, 2023; Yousif et al., 2023; Hasan et al., 2024). Unlike work that establishes image-only prediction, our comparison asks whether adding images improves prediction relative to the analyst's existing prediction model. It follows the predictive distinction emphasized by Mullainathan and Spiess (2017) and complements work using machine learning for hypothesis generation (Ludwig and Mullainathan, 2024).

Choosing a defensible image representation is a prerequisite. Image encoders convert images into numerical features, and plausible choices of encoder can yield materially different results (Nordstrom et al., 2026). We compare modern image representations and store the computed features for reuse before comparing X, I, and $X + I$ models. The companion paper systematically studies how the choice of representation affects prediction. This paper focuses on whether images improve prediction conditional on structured covariates, their predictive overlap with those covariates, the flexibility of the structured model, and how the two sources of information are combined in economic applications.

The image scores are predictions from models estimated using the images; these predictions then enter the second-stage model. They are therefore generated covariates, which makes the separation of training, validation, and test samples important. Cross-fitting prevents an observation's outcome from training its own score, but it does not remove measurement error or make a second-stage coefficient structural (Pagan, 1984; Murphy and Topel, 1985; Fong and Tyler, 2021; Christensen and Hansen, 2026; Battaglia et al., 2024). Related work develops valid inference when AI-generated variables or labels enter regressions (Christensen et al., 2026). Our main object differs: we observe Y in the analysis data and compare out-of-sample prediction losses for estimated prediction methods using X, I, and $X+I$. The predictive-overlap regressions measure how much of the image-based prediction is already captured by the structured covariates. The empirical design has two parts, which affects how readers should interpret the tables. For each application, the headline comparison uses the best-performing structured model and an image prediction method, both selected using validation data. It considers five prespecified ways of combining structured and image information. Appendix A.11 varies how image and structured information are combined and how flexible the structured model is, while holding the sample, split, and outcome fixed. The two parts use different image inputs by design. For horses, we first summarize the image using a prediction from the image model, and then study how that information can be combined with the structured covariates.

The remainder of the paper is organized as follows. Section 2 develops the framework

for distinguishing image predictability, overlap with structured covariates, and incremental predictive value, and presents the Monte Carlo experiments. Section 3 describes the common empirical design and the three datasets. Sections 4–6 present the applications to Toronto house prices, Thoroughbred workout performance, and Ontario academic salaries. Section 7 examines what information the image models recover, and Section 8 draws lessons across the three applications. Section 9 concludes. Additional specifications, robustness checks, and secondary outcomes are reported in Appendix A.

2 Images as Covariates

2.1 Information sets and predictive value

An image encoder maps a raw image into a numerical feature vector, which we call the image representation. We hold the encoder's parameters fixed and denote its image representation by $Z_i = g(I_i)$. A separate prediction model then uses Z_i to predict the outcome. We write the resulting image prediction as

$$\widehat{Y}_i^I = \widehat{f}_I(Z_i). \tag{2}$$

The structured model produces $\widehat{Y}_i^X = \widehat{f}_X(X_i)$. One integration rule combines the two validation predictions,

$$\widehat{Y}_i^{X+I} = \widehat{a} + \widehat{b}_X \widehat{Y}_i^X + \widehat{b}_I \widehat{Y}_i^I. \tag{3}$$

The weights in equation (3) are estimated from validation observations and then applied once to the untouched test set. This specification combines the predictions from the structured and image models in a simple linear regression. We also allow the relationship between the two predictions and the outcome to be nonlinear by combining them with a small neural network. The random-forest specification keeps the structured covariates X_i and adds a single image-based prediction, allowing the contribution of the image information to vary flexibly with the structured variables. None is an unrestricted estimate of $\mathbb{E}[Y_i \mid X_i, I_i]$, and a zero increment can reflect either a lack of residual image information or a combination rule that does not recover it.

This distinction leads to two separate questions. The first is informational: does I improve prediction beyond X? The second concerns integration: if it does, what way of combining the image and structured data best recovers that information? Equation (3) first summarizes the structured data and the image with separate predictions before combining

them. A joint feature model instead uses the structured covariates and image features together to estimate

$$\widehat{Y}_i^{X+I} = \widehat{f}_{XI}(X_i, Z_i). \tag{4}$$

Summarizing each source with a separate prediction can work well when the image and structured data contribute additively to the outcome. This can lose information when the effect of the image depends on X_i, unless the image score preserves the relevant variation. A flexible joint model retains more image inputs, but it also estimates a higher-dimensional function and can perform worse in finite samples. What matters is which information the image representation and image-based prediction retain, and whether the model combining image and structured information can use it. The Monte Carlo compares these approaches using the same model-selection procedure and does not assume in advance which way of combining the data will work best.

At the population level, let $m_X(X) = \mathbb{E}[Y \mid X]$ and $m_{XI}(X, g(I)) = \mathbb{E}[Y \mid X, g(I)]$. Under squared loss, the richer information set cannot worsen the best possible prediction performance:

$$\mathbb{E}[(Y - m_{XI})^2] \leq \mathbb{E}[(Y - m_X)^2]. \tag{5}$$

This result holds for the best possible predictors, not necessarily for the models used in practice. The paper measures estimated models on finite train, validation, and test samples. The estimated increment may be negative because of imperfect image representations, model specification, or model estimation. It is not a formal lower bound on the population predictive value of the image.

Equation (5) concerns optimal predictors. The difference in population mean squared prediction error between estimated models has two components: the population information gain and the difference in approximation and estimation error. Define $\mathcal{R}(f)$ as the population mean squared prediction error,

$$\mathcal{R}(\widehat{f}_X) - \mathcal{R}(\widehat{f}_{XI}) = \underbrace{\mathbb{E}\big[(m_{XI} - m_X)^2\big]}_{\text{population information gain}} + \underbrace{\mathbb{E}\Big[(\widehat{f}_X - m_X)^2\Big] - \mathbb{E}\Big[(\widehat{f}_{XI} - m_{XI})^2\Big]}_{\text{approximation and estimation difference}} \tag{6}$$

A positive observed increment therefore need not indicate an attribute absent from X. It can also arise because the image makes an attribute already recorded in X_i easier to use. We describe the headline estimate throughout as incremental predictive value relative to a

specified model using the structured covariates, and reserve statements about information sets for the population discussion in this section.

For squared-error prediction, Equation (1) summarizes the incremental predictive value of images for the estimated prediction methods being compared. If the image proxies a nonlinear transformation of X that the estimated model using structured covariates misses, $X + I$ can outperform X even though an unrestricted population predictor could recover the same information from X. This is a functional-form limitation rather than an information limitation. Our random-forest comparison asks whether the empirical increment survives a model using structured covariates that captures substantial nonlinearities and interactions. It does not assume that a forest recovers $\mathbb{E}[Y \mid X]$ perfectly. An equivalent loss measure is

$$\Delta L_I = L_X - L_{X+I}, \tag{7}$$

where L is test mean squared error. A positive ΔR_I^2 or ΔL_I means that adding image information lowers mean squared error on the test sample relative to the chosen structured prediction method. It does not establish that the underlying visual attribute is absent from every possible structured dataset.

Five cases clarify the interpretation. First, the image may contain no target-relevant information. Second, it may reveal target-relevant information that X already records. Third, it may reveal an outcome-relevant attribute omitted from X. Fourth, it may provide a better or independent measurement of a trait recorded only noisily or coarsely in X. The last two cases can both generate positive incremental performance, but their economic interpretations differ. Which case applies is not intrinsic to an image: it depends jointly on the outcome, the contents and quality of X, and the estimated prediction model. A fifth case is not an instance of the other four. The image may carry no information about the outcome on its own, and X may carry none on its own, while the two together are informative. Let X and V be independent and equally likely to be -1 or 1, let the image reveal V, and let $Y = XV + \varepsilon$ with independent mean-zero noise. Then

$$\mathbb{E}[Y \mid X] = \mathbb{E}[Y \mid I] = 0, \qquad \mathbb{E}[Y \mid X, I] = XV. \tag{8}$$

Standalone predictability is therefore neither necessary nor sufficient for incremental value. This example also shows why the combination method matters. The best possible predictions using X alone or I alone are both constant, so combining them as in equation (3) cannot capture the interaction between X and I, however well the weights are estimated.

This example shows how summarizing each data source with a single prediction can discard useful information. The comparison must therefore consider alternative ways of combining image and structured information. These cases concern predictive overlap, not the economic value or social desirability of the visual attributes.

2.2 What the image prediction contains

Incremental fit alone does not show what the image score measures. We construct each observation's image-based prediction using a model estimated without that observation's outcome. Alongside equation (1), we then ask how well conventional variables explain these predictions:

$$\widehat{Y}_i^I = \alpha + X_i'\gamma + u_i. \tag{9}$$

A high out-of-sample R^2 in equation (9) means that the estimated model using X_i predicts much of the variation in the image-based prediction. We call this diagnostic *predictive overlap*. Predictive overlap is specific to the estimated image and structured models; it is not a population measure of how much information the raw image and X share. High predictive overlap can coexist with a useful residual component, while low predictive overlap can reflect nuisance variation unrelated to Y_i. Equation (9) measures predictive overlap; equation (1) asks whether adding image information improves prediction of the outcome relative to the estimated structured model. Their joint interpretation is the paper's main empirical device.

We also estimate equation (9) separately for ResNet50, DINOv2, and SigLIP 2 image predictions. Stronger representations can improve prediction for two reasons. They may more accurately reconstruct information already known in X, or they may recover residual visual information that is useful for Y. Appendix A.8 reports these encoder-specific decompositions. The main text emphasizes the validation-selected image model because the economic question is whether the best defensible use of the image contributes beyond X.

2.3 Monte Carlo design

The Monte Carlo holds the joint distribution of image and structured inputs fixed while changing only the outcome equation. Each 224×224 image has a white background and one centered, constant-area superellipse. Two independent standard normal latent variables, S_i and C_i, determine the image's shape and color, respectively. The shape latent S_i changes

the exponent in

$$\left|\frac{x}{a}\right|^n + \left|\frac{y}{a}\right|^n = 1, \tag{10}$$

from circle-like to square-like; the radius a adjusts analytically to preserve area. The color latent C_i follows a one-to-one red–blue–green path, with blue at its midpoint. Area, center, canvas, and background do not vary with either latent.

Two structured covariates partially overlap with the visual latents:

$$X_{1i} = .5S_i + \sqrt{.75}\eta_{1i}, \qquad X_{2i} = .5C_i + \sqrt{.75}\eta_{2i}, \tag{11}$$

where $S_i, C_i, \eta_{1i}, \eta_{2i}$, and ϵ_i are mutually independent standard normal variables. The four DGPs are

$$\begin{aligned}
Y_i^{(1)} &= X_{1i} + X_{2i} + \epsilon_i, \\
Y_i^{(2)} &= X_{1i} + X_{2i} + S_i + C_i + \epsilon_i, \\
Y_i^{(3)} &= X_{1i} + X_{2i} + S_i + C_i + S_iC_i + \epsilon_i, \\
Y_i^{(4)} &= X_{1i} + X_{2i} + 6D_i + \epsilon_i, \qquad D_i = \mathbf{1}\{S_i > s^*,\ |C_i| < 1,\ X_{2i} < 0\},
\end{aligned} \tag{12}$$

with $s^* = \Phi^{-1}(2/3)$. Thus images vary in every case. Images are proxies for X but have no direct outcome term in Case 1. Shape and color enter additively in Case 2 and also interact in Case 3. In Case 4, their contribution depends on a threshold rule involving both image characteristics and a structured covariate. About 11 percent of observations satisfy the Case-4 rule, so that $D_i = 1$. Cases 2–4 have broadly similar signal variances; Table 7 in Appendix A.7 reports the decomposition.

Figure 2 displays the image grid and DGP family. We construct an image for each combination of 101 shape values and 101 color values, then compute and store its features once using the fixed DINOv2 encoder. Each continuous pair (S_i, C_i) is assigned to the nearest point on this grid. The 10,201-state discretization is common to every DGP. Out-of-sample checks show that models using the image features predict both latent variables well. Appendix A.7 uses oracle models—benchmarks that observe X and the true shape and color variables—to separate prediction losses due to the image representation from those due to the model's functional form.

Each of 5,000 replications draws 600 observations and uses a 70/15/15 train/validation/test split. The prespecified models include linear, neural-network, and random-forest (RF) models using X, and linear and neural-network models using image features. To combine the

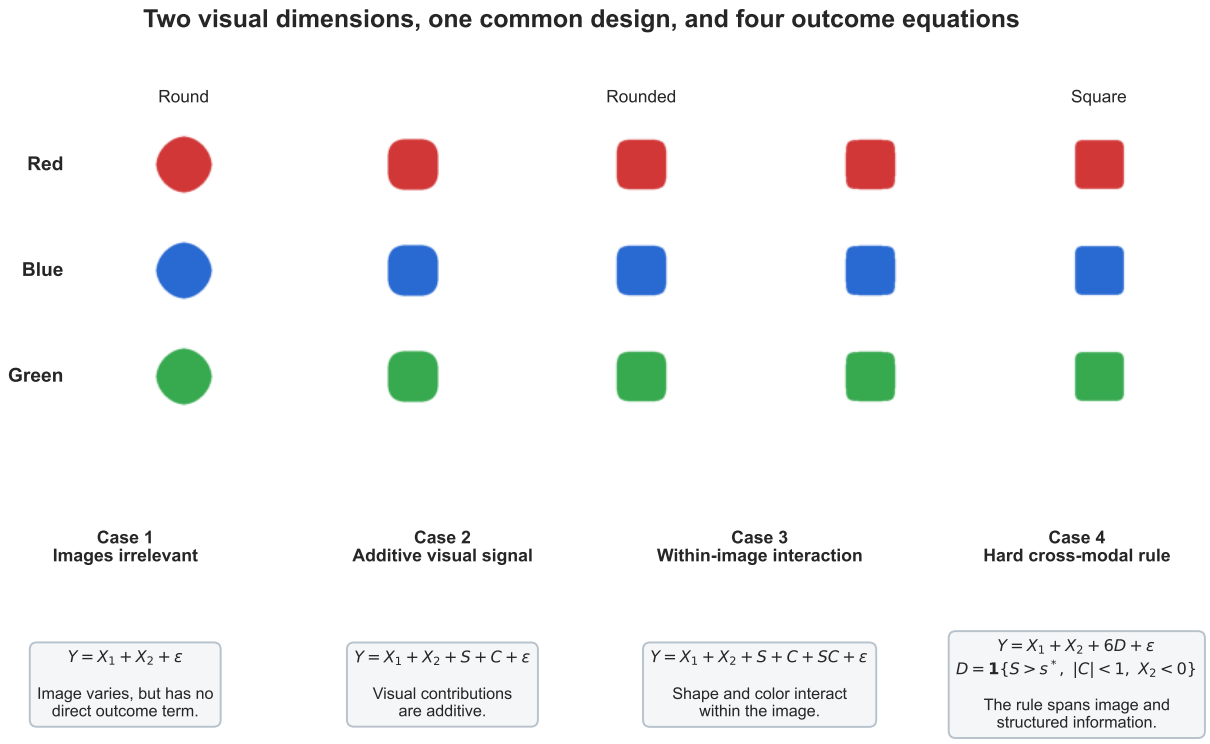


Figure 2: **Synthetic images and the four outcome equations.** Rows in the image grid vary color from red through blue to green; columns vary the constant-area superellipse from round to square. The joint distribution of (X_1, X_2, S, C), the image grid, the disturbance, the sample size and split, and the models considered are identical across cases. Only the displayed outcome relationship changes.

two sources of information, we consider linear and neural-network models using separate predictions, an RF using $(X, \widehat{Y}^I)$, and linear and neural-network models using (X, Z_I) together. We use only validation outcomes to choose the ridge penalty, the number of principal components, and the number and sizes of layers in the small neural networks. We consider the same candidate values and specifications in every case. A pilot study with 100 replications per DGP selects one common random-forest specification for all cases. The main experiment keeps this specification fixed and uses 500 trees. The linear and neural models that combine predictions use only $(\widehat{Y}^X, \widehat{Y}^I)$. The RF model uses X and one cross-fitted image prediction, while the joint models use X and image features reduced by principal components. Linear and neural oracles observe (X_1, X_2, S, C), while the correctly specified oracles in Appendix A.7 receive $S_i C_i$ or D_i.

Figure 3 reports all five integration procedures. The increments for linear-score, NN-score, joint-linear, and joint-NN models are measured against a structured model selected from linear and neural-network candidates using validation data. The RF-score increment is measured against a random forest using X. This preserves the comparison specified in advance for each procedure and avoids retrospectively selecting a structured model with test outcomes. Absolute R^2 levels and the separate structured procedures appear in Appendix A.7.

The results separate the information and integration questions. In Case 1, every image integration method lowers mean held-out R^2; RF score integration lowers it by 0.036 relative to RF X. In Case 2, several procedures recover useful additive information. The joint linear model reaches 0.829, compared with 0.801 for linear score combination and 0.773 for RF score integration. In Case 3, the joint linear model has a mean test R^2 of 0.816, compared with 0.796 for the joint NN and 0.759 for RF score integration. A relationship that is nonlinear in the underlying image attributes can be captured approximately by a linear model using DINOv2 features.

The ranking of the combination methods changes in Case 4. RF score integration reaches mean test $R^2 = 0.610$, compared with 0.485 for the joint NN and 0.461 for linear score combination. Its increment over RF X is 0.282. The forest uses X and a single out-of-sample image-based prediction, yet captures the threshold rule better than the joint neural network using reduced image features. Thus, a complex outcome equation does not necessarily require retaining the full image representation or using a joint neural network. In this design, the image-based prediction retains enough information about shape and color for the random forest to use it together with X_2. Figure 9 in Appendix A.7 summarizes the RF performance

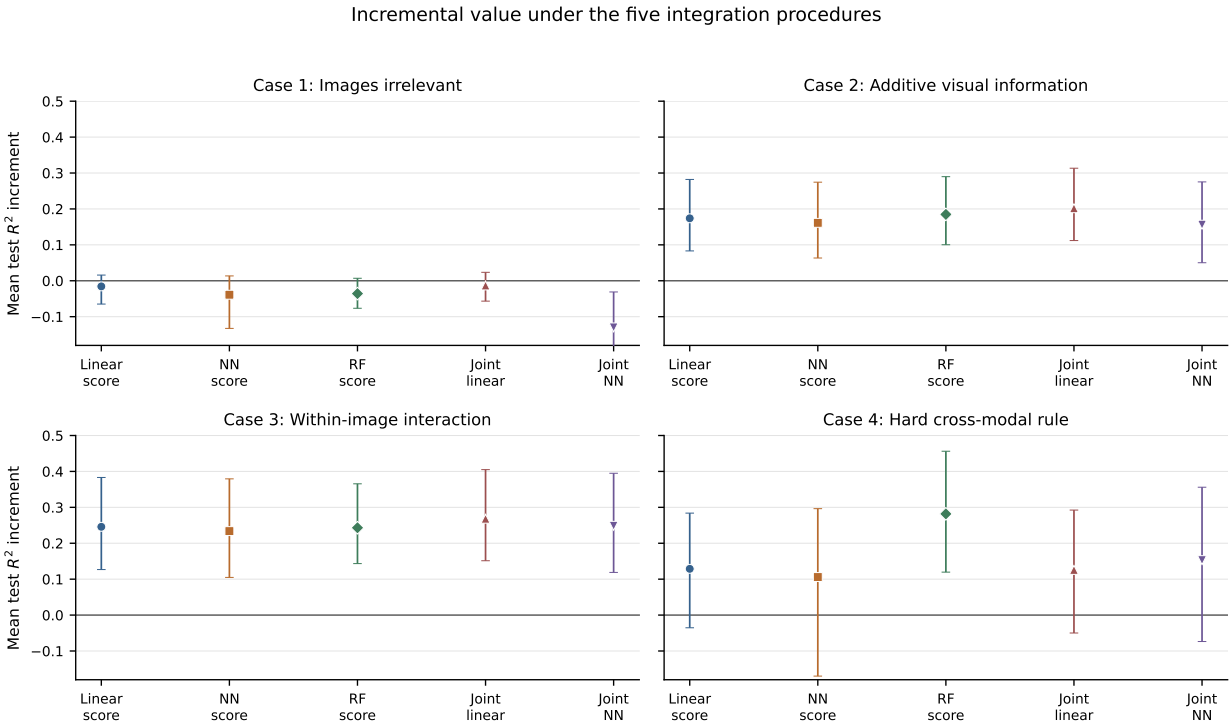


Figure 3: **Incremental value under the five integration procedures.** Each point reports mean $R^2_{X+I} - R^2_X$ over 5,000 replications; whiskers span the 2.5th–97.5th percentiles. For linear score, NN score, joint linear, and joint NN, the benchmark is a model using X, selected from linear and neural-network candidates using validation data. RF score uses X and one image-based prediction from a model estimated without the observation's outcome. Its increment is measured relative to a random forest using X. Every procedure uses the same cached DINOv2 grid, sample, split, and validation-only tuning protocol. Test outcomes are not used to select the structured benchmark.

distributions. This is a finite-sample result for the estimated prediction model. It does not establish that a single image-based prediction generally retains all the image information useful for predicting the outcome given X.

A fifth DGP isolates measurement quality. It uses the same shape and color latents and image grid, but interprets the structured variables as noisy measurements:

$$X_{1i} = \rho S_i + \sqrt{1-\rho^2}\eta_{1i}, \qquad X_{2i} = \rho C_i + \sqrt{1-\rho^2}\eta_{2i}, \qquad Y_i = S_i + C_i + \epsilon_i. \tag{13}$$

The main design sets $\rho = 0.5$. The photograph and X measure the same underlying traits, but their measurement errors differ. Across 5,000 replications, mean R^2 is 0.149 for linear X, 0.139 for NN X, and 0.112 for RF X. It is 0.545 for RF score integration, 0.602 for image-only prediction, 0.604 for the fixed joint linear model, and 0.593 when validation data select among the linear-score, NN-score, joint-linear, and joint-NN models. Flexible modeling of X does not eliminate image value because the photograph supplies another measurement of the latent traits. Positive image value need not imply that the structured data omit the concept entirely. It can arise because the recorded measure is noisy or coarse. Appendix A.17 varies ρ from 0.3 to 0.9; image value declines as the structured measurement becomes more reliable.

The first case retains the earlier caution. Adding irrelevant image inputs cannot worsen the best possible population prediction, but an estimated image-only or joint model can fit variation that merely proxies X or reflects noise. The remaining cases separate two questions: whether images add information beyond X, and how best to use that information for prediction. Additive and within-image relationships favour the joint linear model in this implementation. The threshold rule involving image and structured information favours a random forest that uses X and a single image-based prediction. Performance therefore depends on which image and structured inputs the model retains and how its functional form uses those inputs. This is simulation evidence about one controlled family, not a theorem that any integration architecture dominates. Appendix A.7 reports performance summaries, oracle results, and sensitivity to sample size. It also describes a check using three random initializations. Appendix A.7 also reports an experiment that varies the overlap between image and structured information while holding the functional form fixed.

3 Image Construction and Empirical Design

3.1 Common prediction design

The three applications use the same sequence and the same 12 prespecified group-aware 70/15/15 partitions. We first define the conventional structured information set X, excluding the outcome and variables unavailable at prediction time. Within each split, we hold the image encoder's parameters fixed and reuse its previously computed image features. A small neural network (a multilayer perceptron) predicts the outcome from the image features; ridge regression on these features provides a robustness check. We choose the neural-network specification and regularization settings using only prediction loss on the validation sample.

The preferred image model may use one encoder or a validation-selected combination. Choosing a defensible representation is a prerequisite because encoder choice alone can materially change held-out performance (Nordstrom et al., 2026). For houses, a neural network uses image features combined from several modern encoders. For horses, we combine predictions from neural-network models using different image encoders. For salaries, we use the Ontario image-based prediction supported by the validation results. Appendix A.6 compares encoders using the same neural-network and ridge-regression methods. The companion paper systematically compares image representations. Here, we ask whether adding image information through the selected prediction method improves prediction relative to the estimated model using X.

We choose the primary structured benchmark from prespecified linear, neural-network, and random-forest models using mean squared error on the validation sample. The random forest uses 500 trees. Using validation data, we choose from a common set of candidate values for the fraction of predictors considered at each split, the minimum number of observations in each final group (leaf), and the maximum tree depth. We average predictions from three runs using prespecified random seeds. Continuous inputs use raw values with training-only imputation. For categorical variables with few categories, we retain the existing indicator variables. For training observations in the horse data, we represent variables with many categories using smoothed category-level outcome averages computed from other training folds, keeping observations for the same horse together. Validation data also assess an alternative that represents categories only by their frequencies in the training sample. After choosing the structured model, we use validation data to select how to combine image and structured information. The five candidate methods, fixed before test evaluation, are linear score, NN score, random-forest score, joint linear features, and joint NN features. The random-forest

score model uses X and a single image-based prediction, constructed by cross-fitting that keeps repeated observations for the same unit together. It does not use the image feature vector. Equation (3) shows the linear combination of structured and image predictions; equation (4) describes models that use structured covariates and image features together. We do not use test outcomes to choose image encoders, models using image features, combinations of features or predictions from multiple encoders, structured specifications, or methods for combining image and structured information. We use training data to estimate scaling parameters, imputation rules, principal components, and category encodings. Validation performance determines the number of retained components and when to stop neural-network training. Duplicate images and repeated economic units remain in the same split.

The image-only and joint models use features from encoders with fixed parameters. The features are numerous and highly collinear: DINOv2-B/14 produces 768 features per image, and combining features from several encoders produces more, while the application samples contain 2,320 to 9,729 observations. Some regularization is unavoidable. Ridge, the lasso, and principal-component reduction are all shrinkage procedures, and they differ in the basis in which shrinkage is applied. Ridge downweights the least-squares solution smoothly along each principal direction of the feature matrix, most heavily where sample variance is small. Principal-component regression retains the leading components without penalizing their coefficients and sets coefficients on the discarded components to zero (Hastie et al., 2009). Our joint models use retained components as inputs to ridge regression or neural-network models. The lasso instead shrinks coefficients on individual image features and can set some to zero. It is particularly useful when a small subset of features captures the predictive relationship. Pretrained image features need not have individual interpretations, and predictive information may be spread across many of them. Principal components are linear combinations of image features, so they can summarize information spread across many features. Retaining only the leading components can nevertheless discard information useful for predicting the outcome.

Two further considerations apply here. First, we need to reduce the number of image features supplied to the prediction model. The joint specifications in equation (4) estimate a function of (X_i, Z_i) using a linear model or a neural network. Retaining only the leading principal components reduces the number of image inputs for both types of joint model. Second, we select the number of retained components from $\{16, 32\}$ using validation mean squared error, just as we select the ridge penalty and the neural-network specification. We do not use test outcomes for these choices.

Our main results report average predictive performance and its variation across repeated sample splits. We fixed the number of sample splits at 12 to match the horse stability analysis while keeping estimation of all candidate models feasible for houses and salaries. We use the same number of splits and random seeds across applications. For each sample split, we re-estimate the models and repeat model selection using validation data. For comparisons using the original fixed sample split, we take 2,000 bootstrap draws of the test sample, resampling property, horse, or researcher groups as appropriate. Each draw uses the same resampled groups for the X and $X + I$ models and recomputes the difference in R^2. These intervals measure test-sample uncertainty while holding the estimated models and sample split fixed. They do not capture variation from re-estimating the models or changing the sample split.

Three falsification checks examine the roles of image matching, photographic characteristics, and backgrounds in the predictive results. First, we permute the image-based predictions within the fixed validation and test samples and re-estimate the score-combination model. Second, we augment X with an outcome-independent metadata block containing dimensions, aspect ratio, file size per pixel, brightness, contrast, saturation, channel means, sharpness, and exposure shares. Filenames and identifiers are excluded. Third, for horses we use a pretrained semantic segmentation model to retain the horse and replace the background with a common neutral field before re-encoding the image. The mask uses no outcomes. We also report neighborhood-, sire-, consignor-, and institution-clustered resampling, out-of-cluster split sensitivities, and diagnostics for selection into a usable photograph.

3.2 Applications and information sets

Table 1 gives the repeated-split comparison. The house application has a strong structured benchmark and a single exterior listing photograph. The horse application has standardized side-view conformation photographs and a structured catalogue that is informative but incomplete for physical performance. The salary application uses public profile photographs and a structured benchmark containing research output, citation-history length, and institution.

The outcomes differ in scale, so we emphasize within-application changes. House prices use the logarithm of observed sale price. Horse performance uses speed standardized within sale, which makes workouts conducted at different sales and distances more comparable. Academic salary is the logarithm of the maximum observed disclosed salary in 2024 or 2025; a missing disclosure year is not coded as zero or at the reporting threshold. Tables 2 and 3

Table 1: Structured, image, and combined prediction across 12 repeated splits

Application	N	R^2_X	R^2_I	R^2_{X+I}	ΔR^2_I	Δ as % of R^2_X	% of residual variance	Split range / positive
House prices	7,273	0.890	0.677	0.897	+0.007	+0.8	+6.3	$[-0.007, +0.022]$ / 75%
Horse performance	9,729	0.084	0.044	0.108	+0.024	+28.3	+2.6	$[+0.001, +0.040]$ / 92%
Academic salaries	2,320	0.420	0.102	0.418	−0.002	−0.4	−0.3	$[-0.036, +0.018]$ / 67%

Notes: Entries are means over the same 12 prespecified group-respecting 70/15/15 splits in each application. In each split, validation MSE selects X from linear, neural-network, and random-forest procedures and $X + I$ from linear-score, neural-score, random-forest-score, joint-linear, and joint-neural procedures. Relative MSE reduction is $\Delta R^2_I/(1 - R^2_X)$, the proportional reduction in test mean squared error; because the structured benchmarks leave different amounts of variance to explain, the two scalings order the applications differently. The range is the empirical 2.5th–97.5th percentile across splits, not a confidence interval. Positive is the fraction of splits with $\Delta R^2_I > 0$.

report MSE, MAE, correlation, and calibration slope alongside R^2 for the house and horse applications.

3.3 Generated image scores

An observation's outcome must not help construct an image score later used as a regressor. The main test predictions are out of sample because test outcomes are excluded from estimation of the image prediction model. For descriptive regressions involving observations outside the test set, we create group-cross-fitted image scores. Within each fold, the encoder remains frozen, feature scaling is estimated outside the held-out fold, and the downstream head is trained without that fold's outcomes. The Ontario application uses five folds and averages three fixed seeds for these scores. The house and horse predictive-overlap regressions use validation predictions to fit the X mapping and report performance on test image scores.

Cross-fitting solves a specific leakage problem: an observation's outcome cannot help fit its own generated score. It does not make the score an error-free measurement of a latent visual attribute, remove first-stage estimation dependence, or justify a structural interpretation of its coefficient. These are the familiar generated-regressor and machine-learning-covariate concerns in Pagan (1984), Murphy and Topel (1985), Fong and Tyler (2021), and Christensen and Hansen (2026). We therefore use score coefficients only as descriptive predictive-overlap diagnostics. The headline object is a paired held-out prediction difference, for which the generated score is part of the fitted prediction rule.

4 Toronto House Prices

4.1 Sample and covariates

The house application updates the data and design from *Using Images as Covariates: Measuring Curb Appeal with Deep Learning*. The sample contains 7,273 detached Toronto properties with a sale price, one matched exterior photograph, and split metadata. Repeated addresses and exact duplicate images remain in the same partition.

The outcome is log sale price. The structured design has 231 variables recovered from the historical code: bedrooms, rooms, washrooms, kitchens, lot frontage, geographic coordinates, family rooms, building description, basement, garage, air conditioning, heating, neighbourhood, month, year, and contract date. Training-only medians handle sparse missing numeric values. List price, closing date, and the outcome are excluded. Because X includes 133 neighbourhood indicators and geographic coordinates, the image gain cannot be attributed simply to omitted location.

The exterior photograph can still contain features that MLS variables do not fully record. Figure 4 preserves the earlier paper's motivating example: homes with similar conventional entries can differ in facade, landscaping, materials, apparent upkeep, and relationship to the street. We use these photographs for prediction, not to assign a causal "curb appeal" effect to any one attribute.

4.2 Results

Across 12 repeated partitions, validation selects the random forest in 11 and the NN in one. Mean test R^2 is 0.890 for X, 0.677 for images, and 0.897 for $X + I$. The mean increment is 0.007 (SD 0.010), its empirical 2.5th–97.5th percentile range is $[-0.007, 0.022]$, and 9 of 12 splits are positive. The photograph supplies a modest average improvement, but the sign is less stable than the original locked example suggests.

On the original locked split, validation selects the forest, with test $R^2 = 0.8904$, MSE 0.0247, and MAE 0.1144. The preferred image-only feature-union NN attains $R^2 = 0.6562$. Validation selects joint NN integration, which raises R^2 to 0.9035 and lowers MSE to 0.0217. The locked increment is 0.0130, with a paired property-group bootstrap interval of $[0.0021, 0.0245]$ conditional on the fitted models and split. Images reduce MSE by 11.9 percent.

Log-price RMSE falls from 0.157 to 0.147. Using $\sqrt{\exp(\text{MSE}) - 1}$ as a descriptive con-

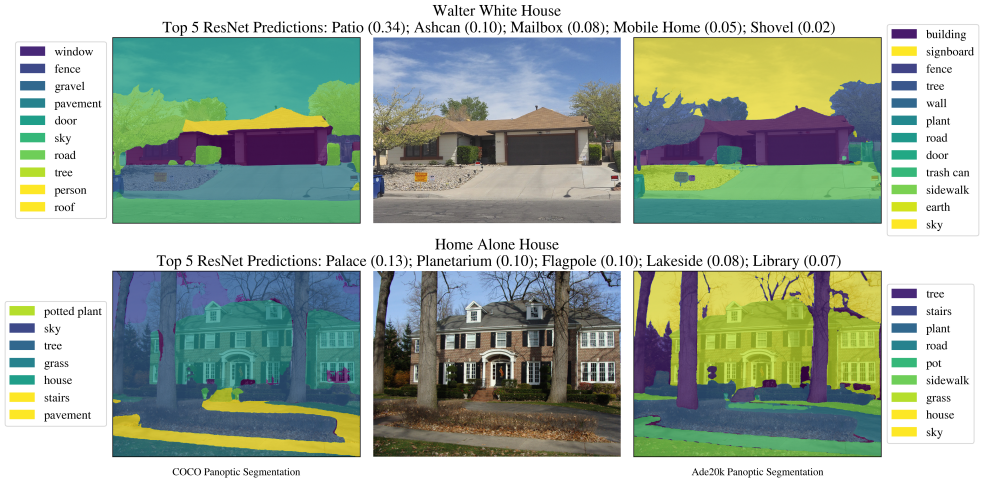


Figure 4: **Exterior photographs contain attributes not fully represented in MLS variables.** The image from the earlier curb-appeal project illustrates visible differences among Toronto properties. The present analysis uses common group-respecting partitions and modern frozen representations; it does not interpret a visual difference as a causal price effect. Note that since the photos from the author pipelines cannot be shared, houses from popular culture have been included as examples. The included images are from Wikimedia Commons and are attributed to Gilliland (2015) and anarchosyn (2009), respectively.

Table 2: Houses: structured, image, and combined prediction

Model	R2	MSE	MAE	Correlation	Calibration slope
Structured only	0.890	0.025	0.114	0.944	1.027
Image only	0.656	0.077	0.206	0.812	0.946
Structured + image	0.903	0.022	0.111	0.951	1.006

Notes: All metrics use the locked test set. Validation chooses among the prespecified linear, NN, and random-forest structured models and among five prespecified integration procedures. The image specification uses cached frozen representations and validation-locked NN heads.

version, approximate percentage-price RMSE falls from 15.8 to 14.8 percent. At the locked test-set median sale price of $1.233 million, this corresponds to an approximate RMSE reduction of $12,100. These translations describe prediction error, not welfare or causal value.

The forest absorbs about half of the increment measured against the structured NN, showing that structured functional form matters, but it does not absorb the entire locked gain. A boosted-tree sensitivity similarly attains $R^2 = 0.892$ from X and 0.903 after adding the image score. The image does not replace MLS data: its error is more than twice the structured error. It improves predictions only at the margin relative to a flexible procedure using rich conventional information.

Outcome-independent image metadata and score permutation address different artifacts. Adding dimensions, file size, brightness, contrast, saturation, channel means, sharpness, and exposure shares to the earlier structured benchmark leaves a positive increment; permuting the fitted image score collapses the gain to zero. DINOv2 alone is slightly stronger image-only than the preferred feature union, yet the union adds more conditional information. Image-only rank and conditional incremental value therefore need not coincide.

The archived MLS records do not contain a consistently populated dedicated living-area variable. A conservative extraction from archived listing descriptions, requiring an explicit 500–15,000 square-foot statement, observed lot frontage, and a one-to-one address match, yields only four qualifying records in the main sample. A meaningful complete-case size sensitivity is therefore not estimable; Appendix A.2 documents this limitation. The headline X nonetheless contains several property-scale measures, including rooms, bedrooms, washrooms, lot frontage, and building/storey categories.

The application connects to work using images to measure urban appearance and property value. Glaeser et al. (2018) and Dubey et al. (2016) emphasize computer vision as a way to measure otherwise opaque features of cities. Law et al. (2019) combines street-view and satellite imagery with property data, while recent work uses self-supervised and multimodal representations for valuation (Yazdani and Raissi, 2023; Yousif et al., 2023; Hasan et al., 2024). Our contribution is the explicit repeated-split $X/I/X + I$ comparison rather than image prediction alone.

5 Thoroughbred Workout Performance

5.1 Sample and outcome

The horse application links 9,729 validated conformation photographs from Ocala Breeders' Sales auctions to under-tack workout performance. The photographs are standardized side views taken before sale. The outcome is `speed_z_within_sale`: metres per second standardized within an OBS sale. This makes workouts at different sales and distances more comparable. Pedigree identities, repeated horses, and duplicate photographs cannot cross partitions.

The structured model uses only observed catalogue and results variables: sex, color, sire, dam, dam sire, consignor, sale, session, state or foaling area, workout-distance source, sale year, hip number, workout distance, and foaling year. High-cardinality variables use group-cross-fitted target encodings and training-frequency encodings. Numeric variables use training medians and missing indicators. We do not observe direct measures of height, weight, leg angles, or conformation.

These omissions make the application a useful incomplete-X case. The photograph can reveal body proportions and visible conformation that the catalogue does not record. Yet it is useful only if a frozen representation and fitted head recover those cues. The preferred score combines validation predictions from modern frozen encoders; all selection predates test evaluation.

5.2 Results

Across 12 repeated partitions, mean test R^2 is 0.084 for validation-selected X, 0.044 for images, and 0.108 for $X + I$. The mean increment is 0.024 (SD 0.012), with empirical 2.5th–97.5th percentiles $[0.001, 0.040]$; 11 of 12 increments are positive. Validation selects linear X in two splits, the NN in eight, and the forest in two. This repeated result leads our interpretation: the sign is stable, but the magnitude is partition sensitive.

On the original locked split, validation selects the structured NN at $R^2 = 0.0983$; the forest attains 0.0933. The image score alone attains 0.1257. The selected NN score combination reaches 0.1896 and lowers MSE from 0.8380 to 0.7532. Its increment of 0.0913 has a paired horse-group bootstrap interval of $[0.0688, 0.1151]$, conditional on this fitted split. This unusually favourable locked value is not representative of the repeated-split distribution.

The result does not say that conformation causally determines performance. Photographs

Table 3: Horses: structured, image, and combined prediction

Model	R2	MSE	MAE	Correlation	Calibration slope
Structured only	0.098	0.838	0.722	0.322	1.228
Image only	0.126	0.813	0.708	0.356	0.935
Structured + image	0.190	0.753	0.686	0.438	1.121

Notes: All metrics use the locked test set. Validation chooses among the prespecified linear, NN, and random-forest structured models and among five prespecified integration procedures. The image specification uses frozen representations and validation-locked NN heads.

may encode age, preparation, grooming, stance, sale practices, background, and camera technology as well as physical form. It says that the implemented pre-sale image procedure predicts subsequent workout speed beyond the fitted pedigree and catalogue procedure.

We replace photographed backgrounds with a common gray field using an outcome-independent horse segmentation model. The masked image alone attains locked $R^2 = 0.1157$, and its selected combination with X reaches 0.1928. Sale-year prediction falls from 0.600 to 0.404. Backgrounds encode collection vintage, but removing them does not remove the speed increment. Adding image metadata to X also leaves a positive locked increment, whereas permuting the image score collapses it. Details of these image-content checks appear in Appendix A.14.

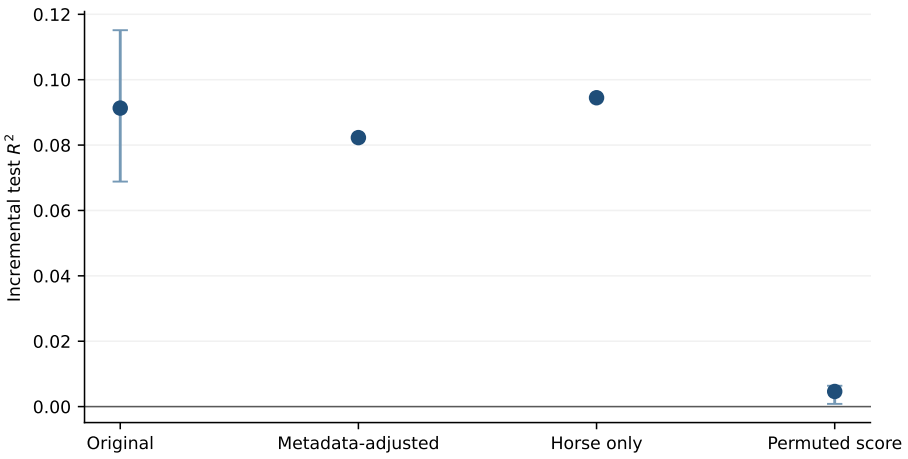


Figure 5: **Horse image falsifications on the locked split.** Original and masked-horse points use locked validation-selected procedures. Metadata adjustment adds outcome-independent image metadata before the image score; permutation breaks the score–outcome match. The whiskers for the original comparison show a grouped bootstrap interval based on the fixed test sample; those for the permuted scores show empirical percentiles. Metadata-adjusted and masked-horse results are point estimates. Appendix A.14 describes these checks.

5.3 Auction-price evidence

Auction prices provide secondary evidence about what the image score represents. Images alone predict log sale price, but a price-trained image score adds little to a strong structured price model once actual speed and catalogue information are included. Conversely, adding image-predicted speed to a held-out speed model containing log price and catalogue variables raises R^2 by 0.0488. The image therefore contains performance information that auction price does not fully summarize. Generated-score coefficients are reported descriptively without conventional second-stage intervals.

Horse racing and wagering markets have long served economists as transparent settings for information aggregation, pricing, and decisions under uncertainty (Thaler and Ziemba, 1988; Sauer, 1998; Snowberg and Wolfers, 2010). Workout speed is relevant to valuation and racing prospects, but whether image signals could improve betting decisions is outside our scope. The comparisons here are predictive, not estimates of how changing appearance would affect speed or price.

6 Ontario Academic Salaries

6.1 Sample and outcome

The third application asks whether public academic profile photographs predict annual salary beyond conventional measures of productivity, career length, and employer. Academic compensation is related to research output, citations, reputation, and seniority (Hamermesh et al., 1982; Moore et al., 2001). Appearance can also predict labour-market outcomes for reasons that are socially mediated and potentially discriminatory (Hamermesh and Biddle, 1994). We therefore treat the photograph strictly as a predictive input and do not assign a normative or causal interpretation to any signal it contains.

The sample contains 2,320 Ontario university researchers with a usable, non-placeholder Google Scholar profile image and a credibly matched Ontario Public Sector Salary Disclosure record in 2024 or 2025. The outcome is

$$Y_i = \log\{\max(Salary_{i,2024}, Salary_{i,2025})\},$$

where the maximum is over observed disclosed salaries. Taxable benefits are excluded. A missing year is not imputed as zero or at the disclosure threshold. Taking the maximum

reduces sensitivity to sabbatical, parental-leave, and partial-year payments. The mean salary is $195,062 and the median is $188,187.

The primary structured information set contains log(1+citations), citation-history length, and institution. Citation-history length is the span between the first and last observed annual citation entries; it is not described as academic age. The sensitivity analyses in Appendix A.15 add h-index, i10-index, and a title/rank category parsed from the disclosure record. Title is highly salary relevant but is partly an administrative description of the compensated position, so it is not in headline X.

Matching never uses salary to resolve identity. The hierarchy requires an exact normalized name and compatible employer, permits middle-initial tolerance, and uses fuzzy names only when employer agreement is strong. A documented crosswalk reconciles university abbreviations, bilingual names, affiliated hospitals, punctuation, and hyphens. Of 4,744 audited Ontario profiles, 2,320 match in either year and 2,126 in both. The matched sample is more senior and cited than unmatched profiles, so it is not representative of all Ontario academics. The analysis is defined directly for the matched disclosure sample.

Google Scholar's default and generic profile icons are removed using exact hashes, perceptual hashes, dimensions, pixel statistics, and contact-sheet inspection. Researchers, repeated names, and duplicate retained headshots remain in the same partition. The source code has separate Ontario and Canada Research Chair branches. The Ontario Scholar sample used in this application comes from work in progress by Webb and Lehrer. We do not draw observations from that project's CRC-specific sample.

6.2 Results

Across 12 repeated partitions, image-only mean R^2 is 0.102. Profile photographs therefore contain salary-correlated information. Yet validation-selected X attains 0.420, and selected $X+I$ attains 0.418. The mean increment is -0.0018 (SD 0.0170), with empirical 2.5th–97.5th percentiles $[-0.0362, 0.0175]$. Eight of 12 point increments are positive, but the distribution is centered near zero and includes larger losses than gains. Validation selects the random forest for X in nine splits and the NN in three; it selects RF-score integration in six, linear score in five, and joint linear in one. Flexible structured and integration procedures do not produce a stable conditional image gain.

The original locked split gives the same qualitative answer. The preferred image stack attains $R^2 = 0.096$. Basic X attains 0.312, while rich X with h- and i10-index measures

attains 0.318. Adding images to rich X lowers R^2 to 0.306, an increment of -0.012 with paired researcher-group interval $[-0.036, 0.007]$. Adding title/rank raises structured R^2 to 0.461 but leaves the image increment at -0.012. The 2025-only and two-year-mean outcomes also have nonpositive locked increments.

Table 4: Ontario academic salary robustness on the locked split

Outcome	N	Image	Basic X	Rich X	Rich $X + I$
Max observed, 2024–25	2,320	0.096	0.312	0.318	0.306
2025 salary	2,236	0.125	0.299	0.289	0.274
Two-year mean (both years)	2,126	0.144	0.378	0.375	0.362

Notes: All entries are untouched-test R^2. Basic X is log citations, citation-history length, and institution. Rich X adds h- and i10-index measures; job-title/rank is reported separately in the text. These alternatives are discussed in Appendix A.15 and do not replace the repeated-split headline design.

The conclusion is conditional on a selected public-disclosure sample. Ontario's reporting threshold and employment-sector coverage make nonappearance informative, and matching rates differ by institution and career-history variables. These limitations constrain external validity, but they do not manufacture the within-sample contrast: images predict salaries alone and fail to improve a flexible fitted procedure using productivity, citation-history length, and institution. The result is the paper's informative-but-redundant empirical case. Profile images predict citations, but the tested integration procedures do not establish a reliable improvement over a strong structured benchmark. The simple score combination is tightly centered near zero, while the flexible score diagnostic remains compatible with a modest gain.

7 What Do Image Models Recover?

The $X + I$ comparison establishes whether adding image information improves out-of-sample prediction of the outcome. Equation (9) asks how much of the image prediction can itself be predicted from X. We call this *predictive overlap*. Figure 6 places the two diagnostics together.

MLS variables explain 0.595 of the fitted house image score, while the image still provides a modest positive average outcome increment. Catalogue and pedigree variables explain only 0.081 of the horse score, which provides the largest mean increment. Salary X explains about 0.111 of the fitted profile-image score across repeated partitions, but the remaining variation

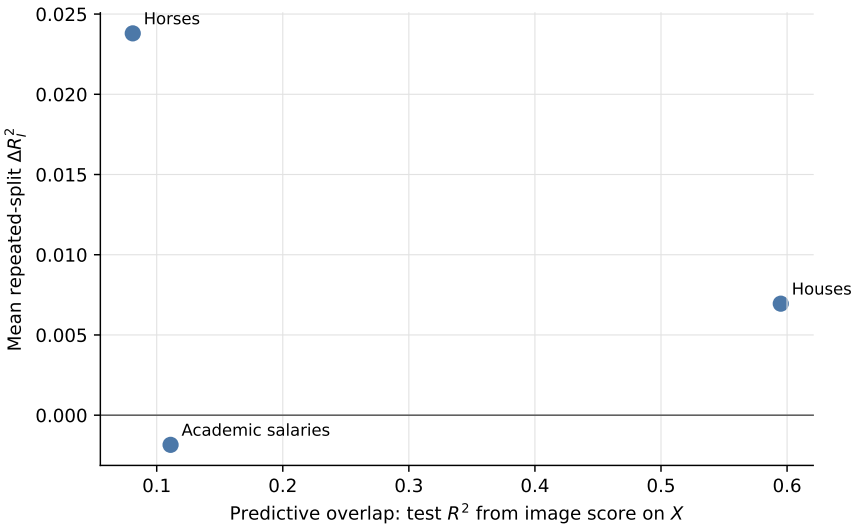


Figure 6: **Predictive overlap and incremental image value.** The horizontal axis is test R^2 from predicting a leakage-safe fitted image score with conventional X. The vertical axis is the mean repeated-split ΔR_I^2 for the outcome. House and horse overlap use the locked validation-to-test diagnostic; salary overlap is averaged over the 12 repeated partitions. The three observations illustrate regimes and do not identify a cross-application relationship.

does not improve salary prediction. Predictive overlap is thus neither a necessary nor sufficient statistic for incremental value. High overlap can leave a useful residual component; low overlap can be nuisance variation unrelated to Y.

The Monte Carlo makes the same point with known visual latents. A fitted image score can have positive predictive overlap in the irrelevant-image DGP because the visual inputs proxy X, even though the image has no direct outcome contribution. Under hard rules, fitted-score overlap can also approximate the mean oracle overlap while tracking it poorly replication by replication. Predictive overlap must therefore be read jointly with held-out ΔR_I^2.

Appendix A.8 compares predictive overlap across image encoders. For houses, MLS variables explain 0.428 of the ResNet image score, 0.620 of the DINOv2 score, and 0.591 of the SigLIP 2 score. Horse overlap remains low across these encoders. For salary, profile images encode career-history and institution-related variation, but that signal is redundant for the outcome after flexible modeling of X.

Direct covariate-prediction exercises make the diagnostic concrete. House representations predict lot frontage, washrooms, bedrooms, coordinates, building type, and garage categories. Horse photographs strongly predict coat colour and predict sale year with R^2 around 0.389 under a common ridge head; background masking reduces, but does not remove, this vintage

signal. Image-derived covariates can therefore encode camera technology, backdrops, digitization, presentation conventions, and institutional processes as well as intrinsic attributes. Profile representations predict citation-history length. Full results appear in Appendix A.9.

8 Cross-Application Synthesis

Incremental value. The repeated design gives three different answers under a common procedure menu. Houses have rich X and strong images; their mean increment is 0.007 and positive in 9 of 12 partitions. Horses have incomplete conventional information about physical form; their mean increment is 0.024 and positive in 11. Academic profile images predict salary alone, but their mean conditional increment is -0.002. Image-only accuracy cannot distinguish these regimes. The three applications cannot be ranked by ΔR_I^2 alone, because their structured benchmarks leave very different amounts of variance to explain. In terms of R^2, the horse increment is roughly three times the house increment. As a share of the prediction error remaining after the structured model, the ordering reverses: houses gain 6.3 percent and horses 2.6 percent across repeated splits, and 11.9 against 10.1 percent on the original locked split. We report both scalings and draw no cross-application ranking from either.

Flexible structured models matter. The locked house increment falls from 0.032 against linear X and 0.027 against NN X to 0.013 against forest X; boosting leaves a similar 0.011 gain. Some apparent image value therefore reflected structured functional form, but a residual locked gain survives two nonlinear models. In the repeated design, however, the average gain relative to validation-selected X is smaller and not positive in every split. Horse images retain a positive mean increment against each structured family, though the locked 0.091 estimate is unusually favourable. Salary's null survives frequent forest selection. These comparisons establish incremental predictive value relative to fitted procedures, not a population theorem that signal is absent from X.

Predictive overlap. The second diagnostic asks how much of the fitted image prediction X can recover. House overlap is high even though the image adds value; horse overlap is low and the increment is positive; salary overlap is also low, but its residual variation is not useful for the target. The two diagnostics answer different questions. Predictive overlap describes the fitted score. Held-out ΔR_I^2 determines whether adding that score or representation helps the fitted outcome procedure.

Integration architecture. If images help, the analyst must still decide how to combine them. The Monte Carlo shows why raw DGP complexity is not a sufficient guide. In Case 3, a nonlinear relationship in raw image attributes becomes approximately linearly accessible after DINOv2 representation. In Case 4, a random forest retaining X and one image score reaches $R^2 = 0.610$, compared with 0.485 for the joint NN. A hard cross-modal rule therefore need not require the complete image representation. What matters is which information survives the representation or score and whether the subsequent prediction model can exploit it.

Empirical integration is similarly application specific. House validation often selects joint NN features, horses select score-level rules, and salaries divide between linear and forest score rules without producing an average outcome gain. Full feature retention is not a default. The appropriate rule is the simplest one supported by validation performance, stability, and deployment cost.

This evidence suggests a sequence for applied work. First, define the intended structured information set and fit a sufficiently flexible benchmark. Second, choose a defensible image representation using validation data; the companion representation-risk paper provides the systematic encoder workflow (Nordstrom et al., 2026). Third, compare X, I, and $X + I$ on common held-out observations, selecting among the candidate integration rules by validation within the same step, and report partition sensitivity. Integration alternatives belong inside this comparison rather than after it: because two modalities can be complementary without either predicting the outcome alone, a single weak combination rule can return a zero increment for an image that does carry conditional information. Fourth, interpret the result with predictive overlap and targeted falsifications. Fifth, if images help, compare score integration, flexible tabular integration, and joint features, then report the simplest validated rule and its cost.

The division of labour between the two papers is deliberate. The companion paper studies representation risk, encoder selection, and encoder combinations across image tasks. This paper treats representation choice as a prerequisite and studies the economic information-set comparison: whether a selected image procedure improves X, what its fitted score overlaps with, and how structured flexibility changes the conclusion. Neither paper claims that one encoder or one integration architecture is universally optimal.

The results also delimit the claims. Images can encode omitted attributes, noisy measurements, transformations missed by the estimated model, or nuisance features and sample-selection processes. We do not call them formal unobservables or interpret fitted scores

causally. The economic value of adding an image depends on the target, the structured information set, the fitted procedures, stability across partitions, and the cost of obtaining and processing the photograph.

9 Conclusion

Image-only prediction is insufficient evidence for adding photographs to an economic dataset. The relevant object is incremental out-of-sample performance relative to a fitted procedure using the analyst's intended structured information set. Across 12 common group-respecting partitions, images raise mean R^2 by 0.007 for Toronto house prices and 0.024 for Thoroughbred workout speed, and lower it by 0.002 for Ontario academic salaries. As a share of the prediction error each structured model leaves, those are gains of 6.3 and 2.6 percent and a loss of 0.3 percent. Profile images predict salaries on their own; their salary-correlated content does not stably improve a validation-selected procedure using citations, citation-history length and institution.

The comparison is procedure relative. Random forests absorb part of the locked house gain, showing that a restricted model using structured covariates can exaggerate image value, but they do not eliminate it in that split. Horse gains remain positive in 11 of 12 repeated selections, although the original locked estimate is unusually large. Partition dispersion therefore belongs beside locked-test bootstrap uncertainty. Predictive overlap supplies interpretation but cannot replace the outcome comparison.

Once images help, integration remains empirical. The Monte Carlo shows that neither nonlinear raw image structure nor a hard cross-modal rule mechanically requires a joint NN or the complete image representation. The recommendation is sequential: begin with a sufficiently flexible model using structured covariates, test whether adding images improves its predictive performance on common held-out observations, and then choose the simplest validation-supported integration rule whose gain justifies its instability and cost. Images can be useful covariates, but their value depends on the outcome, the structured information set, and the fitted procedure used to combine the two.

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A Online Appendix

A.1 Source projects and preservation

The house and horse comparisons reuse the locked samples and splits from the completed representation-risk analyses. That reuse is deliberate: changing the split after observing model performance would make the X / I / $X + I$ comparison partly test-selected. The Ontario sample had no defensible three-way split after the stricter placeholder audit, so one reproducible 70/15/15 split was constructed at the researcher and duplicate-group level and locked before any embedding or outcome modelling. The Ontario copy contains only rows with `source_group == ontario` and no observations from the CRC-specific sample. Directory layout, file hashes, cache locations and audit logs are documented in the replication package.

A.2 House construction and structured specification

The Toronto source image directory and structured data were matched by normalized address in the redesigned house project. The final manifest stores property identifier, address group, image path, SHA-256 hash, outcome, original historical split flags, new locked split, and original structured row index. Exact image hashes and normalized addresses define the leakage groups. The 7,273-property sample is common to every representation.

The 231 structured variables are taken from the saved specification rather than reconstructed from variable names after the fact. They include seven core continuous/count variables (beds, rooms, washrooms, kitchens, lot frontage, longitude, and latitude), a family-room variable, 19 building-description indicators, 14 basement indicators, eight garage indicators, 12 month indicators, 24 year indicators, contract date, five air-conditioning indicators, seven heating indicators, and 133 neighbourhood indicators. The outcome, list price, and closing date are excluded. Scaling and the 14 basement imputations are learned from training observations only.

The old curb-appeal paper used a smaller approximately 6,900-house sample, historical CNN encodings, and a different downstream design. Its motivation and selected house images remain useful, but its headline errors are not treated as exact replications. The present estimates use one common sample, modern and legacy frozen features, and group-respecting partitions.

The source audit found no consistently populated dedicated living-area field in the main

MLS extract. An archived description workbook contains free text and lot frontage for a much smaller historical subset. We extracted the largest explicit 500–15,000 square-foot statement after excluding land- and lot-only contexts, required observed lot frontage, and required a unique normalized-address match to the 7,273-property sample. Only four records qualify. Because this is below the prespecified minimum of 50, no complete-case model is estimated. The audit table and extraction rule are machine readable. List price is not used.

A.3 Horse construction and structured specification

The OBS project constructs one row per horse–sale record with a validated conformation image and normalized workout speed. Image validation rejects unreadable files and implausible dimensions. Horse grouping combines available name, pedigree, and photo-identity evidence. All repeated identities remain within one split.

The structured model includes sex, color, sire, dam, dam sire, consignor, sale, session, state or foaling area, workout-distance source, sale year, hip, workout distance in furlongs, and foaling year parsed from the catalogue field. For each high-cardinality categorical variable, a five-fold horse-group procedure creates a smoothed target encoding in the training sample. Validation and test encodings use the full training map. A log-frequency feature accompanies each target encoding. Numeric variables use training medians and missing indicators. The final specification has no height, weight, veterinary, or manually scored conformation variables.

The outcome, `speed_z_within_sale`, standardizes metres per second within each sale. An alternative standardizes workout time within sale and distance; the main outcome is preferred because it preserves the intuitive direction that larger values indicate better performance while making distances and sales more comparable.

A.4 Ontario academic-salary construction

The source project has separate Ontario Scholar and CRC-specific branches. The preparation script filters to `source_group == ontario` before hashes, outcomes, duplicate groups, or splits are constructed. The JAE application uses the Ontario sample from the Lehrer–Webb project and does not draw observations from its CRC-specific sample. The audit log records source paths and counts and confirms zero CRC-specific rows in the analysis copy.

Official Ontario Public Sector Salary Disclosure CSVs contain 377,881 rows in 2024 and 404,922 in 2025. Files, URLs, download dates, and SHA-256 hashes are stored in the source

inventory. Matching requires normalized name and compatible institution/employer; salary is never used to resolve identity. Exact name–employer matches are preferred, middle-initial tolerance is controlled, and fuzzy names are accepted only under strong employer agreement. Twelve ambiguous person-year candidates were manually rejected and none accepted unresolved. The resulting sample contains 2,320 researchers observed in either year and 2,126 in both.

The primary outcome is log maximum observed 2024–25 salary, excluding benefits. Basic X contains log citations, citation-history length, and institution. Citation-history length is the inclusive span of observed annual citation entries and is not treated as an independent measure of career age. Rich-X, rank/title, 2025-only salary, and two-year-mean salary are sensitivities.

The Google graduation-cap default has SHA-256 hash

`28f06a0cb1e85bfd834635ee195424941d1ccd8d3e557e1b1bed1bc62ce57b76`.

It appears exactly in 2,634 Ontario rows. We also calculate 64-bit average, difference, and perceptual hashes after standardized resizing. Six images share the frequent perceptual signature and icon geometry. A contact sheet of the historical scraper flags showed that the flag was overinclusive: it marked many real portraits because they were small or compressed. We therefore do not remove a row solely for that flag. Visual inspection identifies four additional invalid images: one white, one black, one gray, and one generic outlined-person icon. The machine-readable removal table stores the author ID, source path, all hashes, dimensions, file size, perceptual distance, and reason.

A.5 Frozen representations and NN heads

Modern encoders use the same checkpoints as the representation-risk projects: torchvision ResNet50 with ImageNet-1K V1 weights; DINOv2-B/14 (`facebook/dinov2-base`); SigLIP 2-B/16 (`google/siglip2-base-patch16-224`); torchvision ConvNeXt-B; and torchvision ViT-B/16. Legacy house and horse features include VGG16, InceptionV3, MobileNet, and semantic COCO/ADE20K representations where validated. The encoder is frozen in every specification. The cache stores one pooled feature vector per image, row-order digests, checkpoint metadata, dimension, timing, peak GPU allocation, and SHA-256 checksum.

The common NN head standardizes embeddings using training means and standard deviations, then applies one or two GELU hidden layers with dropout and a scalar output. AdamW minimizes mean squared error. Architecture selection uses a small prespecified grid

on validation observations; it is not performed separately for every encoder. Five seeds (101, 202, 303, 404, and 505) are averaged for headline existing-project NN predictions. The Ontario architecture is selected from 256; 256–64; and 512–128 hidden-unit candidates using mean validation MSE across ResNet and DINOv2 anchors and three seeds, then fixed across the three encoders. Early stopping uses validation loss. Ridge heads use a common logarithmic penalty grid.

A.6 Encoder and downstream-head results

The detailed encoder tables are implementation checks rather than the paper's main result. They establish that the image conclusion is not an accident of one weak encoder. In houses, modern representations are much stronger than common legacy defaults under both ridge and NN heads. In horses, DINOv2 is the strongest modern single model under the locked validation rule, while stacking adds relatively little image-only performance. In salaries, the validation-supported three-encoder score predicts salary alone but does not improve the fitted structured procedure. The companion paper reports the systematic encoder benchmark.

Table 5: Toronto houses: frozen image representations under common heads

Encoder	Ridge R2	NN R2	NN seed SD	NN correlation	NN calibration slope
ResNet50	0.395	0.450	0.002	0.672	0.948
VGG16	0.385	0.400	0.008	0.638	0.884
InceptionV3	0.431	0.452	0.006	0.673	0.950
MobileNetV2	0.431	0.452	0.003	0.676	0.912
COCO semantic	0.152	0.178	0.007	0.429	0.846
ADE20K semantic	0.202	0.251	0.003	0.503	0.919
DINOv2-B/14	0.637	0.665	0.004	0.816	0.985
SigLIP 2-B/16	0.630	0.638	0.005	0.801	0.937
ConvNeXt-B	0.508	0.548	0.004	0.741	0.949
ViT-B/16	0.470	0.503	0.006	0.712	0.925

Notes: Ridge and NN heads use the same locked sample and split. NN predictions average five prespecified seeds; seed SD is across test metrics, and no best test seed is selected.

Table 6: OBS horse speed: frozen image representations under common heads

Encoder	Ridge R2	NN R2	NN seed SD	NN correlation	NN calibration slope
ResNet50	0.043	0.037	0.006	0.203	0.768
VGG16	0.038	0.037	0.003	0.193	0.949
InceptionV3	0.038	0.043	0.005	0.209	0.876
MobileNetV2	0.047	0.044	0.003	0.216	0.819
COCO semantic	0.000	0.002	0.001	0.061	2.372
ADE20K semantic	0.001	0.003	0.006	0.056	0.781
DINOv2-B/14	0.137	0.126	0.006	0.356	0.935
SigLIP 2-B/16	0.036	0.033	0.005	0.189	0.785
ConvNeXt-B	0.040	0.043	0.006	0.222	0.748
ViT-B/16	0.048	0.055	0.005	0.242	0.791

Notes: Ridge and NN heads use the same locked sample and split. NN predictions average five prespecified seeds; seed SD is across test metrics, and no best test seed is selected.

A.7 Monte Carlo details

The area of a symmetric superellipse is

$$A(n, a) = 4a^2 \frac{\Gamma(1 + 1/n)^2}{\Gamma(1 + 2/n)}. \tag{14}$$

For each shape exponent n, the script solves $a = \sqrt{A_0/[4\Gamma(1 + 1/n)^2/\Gamma(1 + 2/n)]}$ with target area $A_0 = 10{,}500$ pixels. Images are 224×224 pixels and centered at $(112, 112)$. The shape latent maps through its normal CDF into exponents from approximately 1.8 to 8.0. The color latent maps through its normal CDF to $t \in (0, 1)$ and then follows a piecewise-linear RGB path from $(210, 55, 55)$ to $(40, 105, 210)$ to $(55, 170, 80)$. The path is one-to-one on either side of its shared blue midpoint. Shape and color each use 101 grid values, for 10,201 images.

The rasterized foreground area ranges from 10,588 to 10,780 pixels across shape states, a 1.83 percent range around the analytical target. It is exactly constant across colors for a given shape. Masks are pixel-identical across all colors, RGB values are identical across shapes at a given color state, and the foreground centroid never differs from the fixed center by more than half a pixel. Filenames and grid indices are not model inputs. Repeated grid states are acceptable because the frozen encoder is computed before outcome construction and carries no fitted outcome information; PCA, scaling, heads, and tuning remain train-only or validation-only.

Table 7 reports a separate two-million-draw calculation of signal components. Cases 2–4 hold all coefficients and the disturbance variance fixed except for the displayed functional-form term. Their total outcome variances are broadly comparable. The Case-4 rule occurs in approximately 11 percent of observations, leaving roughly 48 rule observations in a training sample of 420.

Table 7: Signal variance and Case-4 regime frequency

DGP	Var(X)	Visual additive	Interaction/rule	Total signal	Total outcome	Rule frequency
Case 1	2.000	0.000	0.000	2.000	3.000	N/A
Case 2	2.000	1.999	0.000	5.998	6.998	N/A
Case 3	2.000	1.999	0.998	6.992	7.992	N/A
Case 4	2.000	0.000	3.628	5.384	6.384	0.114

Notes: Entries use a separate two-million-draw design calculation. Component variances do not generally sum to total signal variance because X overlaps with S and C; covariance terms are reported in the machine-readable file.

The common estimator menu is set before DGP-specific test results. Linear specifications use a ridge grid of $\{.01, .1, 1, 10\}$. Image and joint linear models select 16 or 32 train-only DINO principal components. NN image and joint heads use 32 components and select between one 32-unit layer and 64–16 hidden units. Adam, ReLU activations, early stopping, and the remaining optimization settings are identical across DGPs. The linear score model regresses validation outcomes on the structured-NN and image-NN validation predictions. Because there are only 90 validation observations, the small neural-network model combining the predictions uses threefold validation-internal selection over 4 and 8–4 hidden-unit architectures and a common regularization grid, then fits the chosen meta-model on the full validation set. Test outcomes never enter these decisions. One deterministic initialization is used per candidate and replication; a 100-replication subset repeats the NN procedures under three initialization variants.

The forest extension uses the same $N = 600$ observations, splits, outcomes, cached image grid, and 5,000 replications per DGP. A prespecified pilot of 100 replications per DGP compares feature-subsampling, leaf-size, and depth candidates using validation MSE normalized by validation outcome variance. It selects one configuration across all five DGPs, which is then frozen at 500 trees. The structured forest receives (X_1, X_2). The RF score model receives $(X_1, X_2, \widehat{Y}^I)$, where $\widehat{Y}^I$ is a DINOv2 ridge score constructed out of fold for training observations, from training only for validation, and from training plus validation for test. It never receives raw image coordinates. This isolates whether a flexible model that captures interactions among structured covariates can use a scalar image score without retaining the full representation.

Frozen DINOv2 preserves the designed visual dimensions. On 4,500 held-out diagnostic observations, linear and NN heads recover S with R^2 0.975 and 0.990 and recover C with 0.979 and 0.997. The blue-and-square indicator has ROC AUC 0.9994 under both heads. A weak nonlinear result in the outcome experiment therefore cannot be attributed to an invisible color or shape range. At the same time, a linear head on DINO features is not linear in raw pixels: the frozen encoder may already expose nonlinear combinations of shape and color.

Figure 7 reports the key raw performance distributions, and Figure 8 summarizes the original linear/NN procedures. Table 8 gives their complete R^2, MSE, and MAE results. The forest extension follows in Table 13 and Figure 9. Correctly specified Case-3 and Case-4 oracles add the true S_iC_i term or rule indicator; they are diagnostics rather than feasible image estimators.

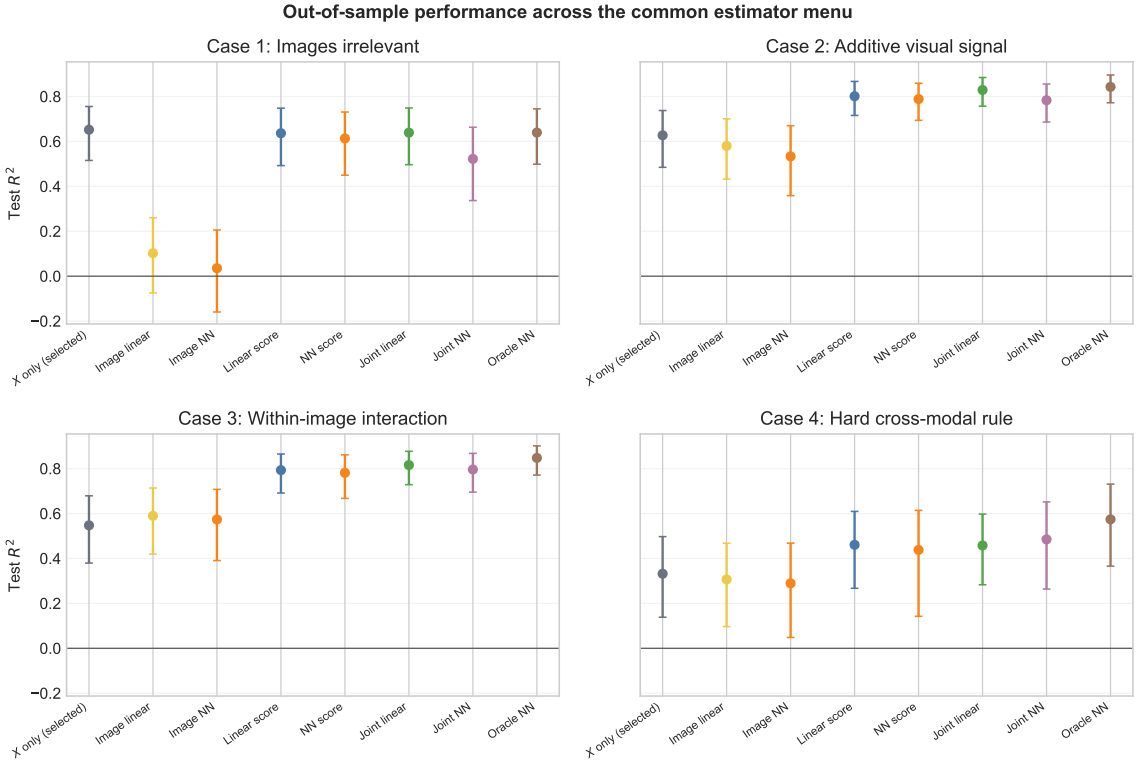


Figure 7: **Raw performance across the four DGPs.** Points report mean untouched-test R^2 over 5,000 replications; whiskers span the 2.5th to 97.5th percentiles. The same samples, cached DINOv2 grid, estimator menu, and validation rule are used in each panel. The oracle NN observes (X_1, X_2, S, C) rather than image embeddings.

Table 8: Four-DGP Monte Carlo: Case 1 (images conditionally redundant)

Procedure	Mean R^2	Median	SD	MCSE	P2.5	P5	P95	P97.5	Mean MSE	Mean MAE
Correct functional-form oracle	0.653	0.659	0.061	0.0009	0.517	0.544	0.742	0.756	1.009	0.802
Image NN	0.035	0.041	0.093	0.0013	-0.159	-0.126	0.179	0.206	2.841	1.344
Image linear	0.102	0.106	0.086	0.0012	-0.075	-0.043	0.236	0.260	2.643	1.297
Joint NN	0.522	0.530	0.085	0.0012	0.337	0.371	0.647	0.663	1.392	0.938
Joint linear	0.639	0.646	0.065	0.0009	0.497	0.525	0.733	0.749	1.050	0.818
Linear score	0.637	0.642	0.065	0.0009	0.492	0.519	0.732	0.748	1.058	0.821
Linear score combination (linear bases)	0.644	0.650	0.064	0.0009	0.501	0.532	0.737	0.752	1.036	0.812
NN score	0.613	0.624	0.085	0.0012	0.449	0.486	0.716	0.731	1.128	0.845
Oracle NN	0.640	0.645	0.064	0.0009	0.499	0.526	0.733	0.745	1.050	0.818
Oracle linear additive	0.653	0.659	0.061	0.0009	0.517	0.544	0.742	0.756	1.009	0.802
X NN	0.646	0.651	0.062	0.0009	0.508	0.536	0.737	0.752	1.030	0.810
X linear	0.655	0.661	0.061	0.0009	0.520	0.547	0.744	0.758	1.005	0.800
X only (selected)	0.652	0.658	0.061	0.0009	0.515	0.542	0.742	0.756	1.013	0.803

Notes: Each row summarizes 5,000 independent samples of 600 observations. All methods use the same 70/15/15 split, tuning grid, and cached frozen DINOv2 image grid. MCSE is the Monte Carlo standard error of the mean.

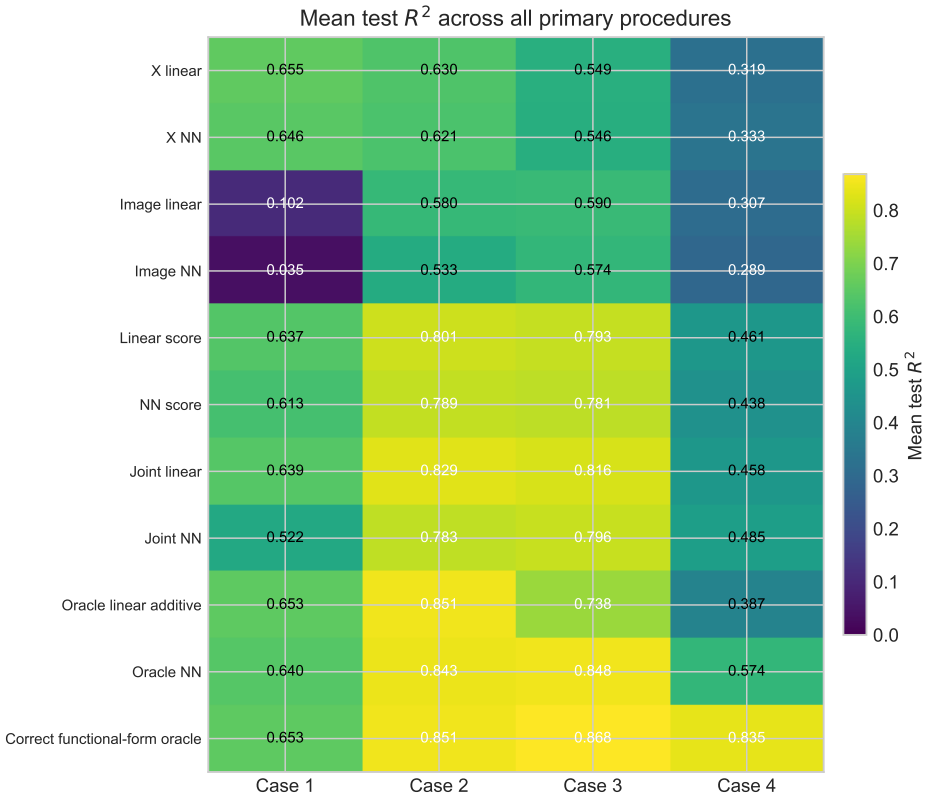


Figure 8: **Mean test R^2 for the original linear/NN estimator menu.** Cells summarize 5,000 replications. The correctly specified oracle receives the true within-image interaction in Case 3 and the true hard-rule indicator in Case 4. Random-forest results appear separately in Table 13 and Figure 9.

Table 9: Four-DGP Monte Carlo: Case 2 (additive visual signal)

Procedure	Mean R^2	Median	SD	MCSE	P2.5	P5	P95	P97.5	Mean MSE	Mean MAE
Correct functional-form oracle	0.851	0.854	0.030	0.0004	0.784	0.797	0.894	0.900	1.009	0.802
Image NN	0.533	0.540	0.079	0.0011	0.359	0.392	0.652	0.670	3.183	1.419
Image linear	0.580	0.586	0.069	0.0010	0.432	0.458	0.685	0.701	2.867	1.349
Joint NN	0.783	0.789	0.044	0.0006	0.686	0.705	0.846	0.856	1.470	0.962
Joint linear	0.829	0.833	0.033	0.0005	0.757	0.770	0.878	0.885	1.159	0.858
Linear score	0.801	0.805	0.039	0.0006	0.716	0.731	0.859	0.867	1.350	0.923
Linear score combination (linear bases)	0.824	0.827	0.035	0.0005	0.747	0.762	0.874	0.881	1.196	0.871
NN score	0.789	0.795	0.066	0.0009	0.694	0.716	0.850	0.859	1.439	0.948
Oracle NN	0.843	0.846	0.032	0.0004	0.772	0.785	0.889	0.896	1.065	0.823
Oracle linear additive	0.851	0.854	0.030	0.0004	0.784	0.797	0.894	0.900	1.009	0.802
X NN	0.621	0.628	0.065	0.0009	0.476	0.507	0.719	0.733	2.579	1.281
X linear	0.630	0.636	0.065	0.0009	0.487	0.515	0.726	0.739	2.520	1.266
X only (selected)	0.627	0.633	0.065	0.0009	0.485	0.513	0.723	0.738	2.539	1.271

Notes: Each row summarizes 5,000 independent samples of 600 observations. All methods use the same 70/15/15 split, tuning grid, and cached frozen DINOv2 image grid. MCSE is the Monte Carlo standard error of the mean.

Table 10: Four-DGP Monte Carlo: Case 3 (within-image interaction)

Procedure	Mean R^2	Median	SD	MCSE	P2.5	P5	P95	P97.5	Mean MSE	Mean MAE
Correct functional-form oracle	0.868	0.872	0.030	0.0004	0.800	0.814	0.910	0.916	1.012	0.803
Image NN	0.574	0.582	0.081	0.0011	0.391	0.428	0.690	0.708	3.303	1.440
Image linear	0.590	0.597	0.075	0.0011	0.420	0.455	0.699	0.714	3.185	1.410
Joint NN	0.796	0.801	0.044	0.0006	0.695	0.716	0.859	0.868	1.575	0.989
Joint linear	0.816	0.821	0.038	0.0005	0.729	0.746	0.870	0.877	1.428	0.934
Linear score	0.793	0.798	0.044	0.0006	0.691	0.715	0.855	0.865	1.602	0.991
Linear score combination (linear bases)	0.795	0.802	0.045	0.0006	0.691	0.713	0.856	0.865	1.596	0.980
NN score	0.781	0.791	0.076	0.0011	0.667	0.698	0.851	0.861	1.697	1.011
Oracle NN	0.848	0.852	0.034	0.0005	0.771	0.786	0.895	0.901	1.173	0.858
Oracle linear additive	0.738	0.745	0.055	0.0008	0.610	0.637	0.814	0.824	2.037	1.090
X NN	0.546	0.552	0.075	0.0011	0.376	0.413	0.659	0.677	3.558	1.435
X linear	0.549	0.555	0.074	0.0010	0.385	0.420	0.658	0.676	3.534	1.432
X only (selected)	0.548	0.554	0.075	0.0011	0.380	0.414	0.659	0.679	3.543	1.433

Notes: Each row summarizes 5,000 independent samples of 600 observations. All methods use the same 70/15/15 split, tuning grid, and cached frozen DINOv2 image grid. MCSE is the Monte Carlo standard error of the mean.

Table 11: Four-DGP Monte Carlo: Case 4 (hard cross-modal rule)

Procedure	Mean R^2	Median	SD	MCSE	P2.5	P5	P95	P97.5	Mean MSE	Mean MAE
Correct functional-form oracle	0.835	0.839	0.036	0.0005	0.752	0.769	0.885	0.893	1.012	0.803
Image NN	0.289	0.299	0.105	0.0015	0.048	0.104	0.444	0.468	4.408	1.655
Image linear	0.307	0.316	0.094	0.0013	0.097	0.141	0.446	0.468	4.303	1.643
Joint NN	0.485	0.494	0.098	0.0014	0.264	0.318	0.629	0.652	3.187	1.368
Joint linear	0.458	0.465	0.081	0.0011	0.283	0.314	0.576	0.598	3.368	1.416
Linear score	0.461	0.468	0.087	0.0012	0.267	0.311	0.587	0.610	3.350	1.400
Linear score combination (linear bases)	0.439	0.446	0.083	0.0012	0.258	0.297	0.557	0.578	3.493	1.434
NN score	0.438	0.458	0.137	0.0019	0.142	0.238	0.589	0.614	3.492	1.399
Oracle NN	0.574	0.585	0.096	0.0014	0.366	0.405	0.713	0.731	2.641	1.210
Oracle linear additive	0.387	0.390	0.072	0.0010	0.233	0.263	0.497	0.515	3.835	1.466
X NN	0.333	0.338	0.092	0.0013	0.137	0.176	0.473	0.500	4.177	1.460
X linear	0.319	0.322	0.083	0.0012	0.141	0.175	0.450	0.471	4.272	1.481
X only (selected)	0.332	0.336	0.091	0.0013	0.138	0.176	0.472	0.497	4.185	1.463

Notes: Each row summarizes 5,000 independent samples of 600 observations. All methods use the same 70/15/15 split, tuning grid, and cached frozen DINOv2 image grid. MCSE is the Monte Carlo standard error of the mean.

Table 12: Incremental R^2 by integration procedure

Case	Procedure	Mean	Median	SD	MCSE	P2.5	P5	P95	P97.5	Pr($\Delta > 0$)
1	Joint NN	-0.130	-0.125	0.056	0.0008	-0.253	-0.229	-0.044	-0.031	0.003
1	Joint linear	-0.013	-0.012	0.020	0.0003	-0.057	-0.048	0.017	0.024	0.237
1	Linear score	-0.016	-0.012	0.021	0.0003	-0.065	-0.054	0.012	0.016	0.205
1	NN score	-0.039	-0.028	0.063	0.0009	-0.133	-0.103	0.008	0.013	0.104
2	Joint NN	0.156	0.154	0.057	0.0008	0.050	0.066	0.252	0.275	0.998
2	Joint linear	0.202	0.199	0.051	0.0007	0.112	0.123	0.292	0.313	1.000
2	Linear score	0.174	0.171	0.050	0.0007	0.083	0.097	0.259	0.282	1.000
2	NN score	0.161	0.162	0.073	0.0010	0.063	0.080	0.253	0.274	0.996
3	Joint NN	0.249	0.245	0.070	0.0010	0.119	0.141	0.369	0.395	1.000
3	Joint linear	0.268	0.264	0.065	0.0009	0.151	0.171	0.380	0.405	1.000
3	Linear score	0.246	0.242	0.065	0.0009	0.127	0.148	0.360	0.383	1.000
3	NN score	0.234	0.236	0.091	0.0013	0.105	0.127	0.356	0.379	0.993
4	Joint NN	0.153	0.156	0.110	0.0016	-0.074	-0.030	0.325	0.356	0.919
4	Joint linear	0.126	0.128	0.086	0.0012	-0.050	-0.019	0.260	0.292	0.926
4	Linear score	0.129	0.129	0.081	0.0011	-0.035	-0.001	0.256	0.284	0.949
4	NN score	0.106	0.120	0.137	0.0019	-0.170	-0.083	0.264	0.296	0.879

Notes: The baseline is the structured-only model selected between linear and NN specifications using validation MSE. The same selected baseline is subtracted from all four integration procedures within a replication. Percentiles describe Monte Carlo distributions.

Table 13: Random forests in the five Monte Carlo DGPs

DGP	Linear X	NN X	RF X	Linear score	NN score	RF X+score	Joint linear	Joint NN
1. Conditionally redundant	0.655	0.646	0.612	0.637	0.613	0.577	0.639	0.522
2. Additive	0.630	0.621	0.588	0.801	0.789	0.773	0.829	0.783
3. Within-image interaction	0.549	0.546	0.516	0.793	0.781	0.759	0.816	0.796
4. Hard cross-modal rule	0.319	0.333	0.328	0.461	0.438	0.610	0.458	0.485
5. Noisy X measurement	0.149	0.139	0.112	0.545	0.523	0.545	0.604	0.540

Notes: Entries are mean untouched-test R^2 over 5,000 replications. A 100-replication-per-DGP pilot selects one RF feature-subsampling, leaf-size, and depth configuration using normalized validation MSE; that configuration and 500 trees are then frozen for every DGP. The RF score model receives X and a leakage-safe scalar DINOv2 ridge prediction, not raw image coordinates.

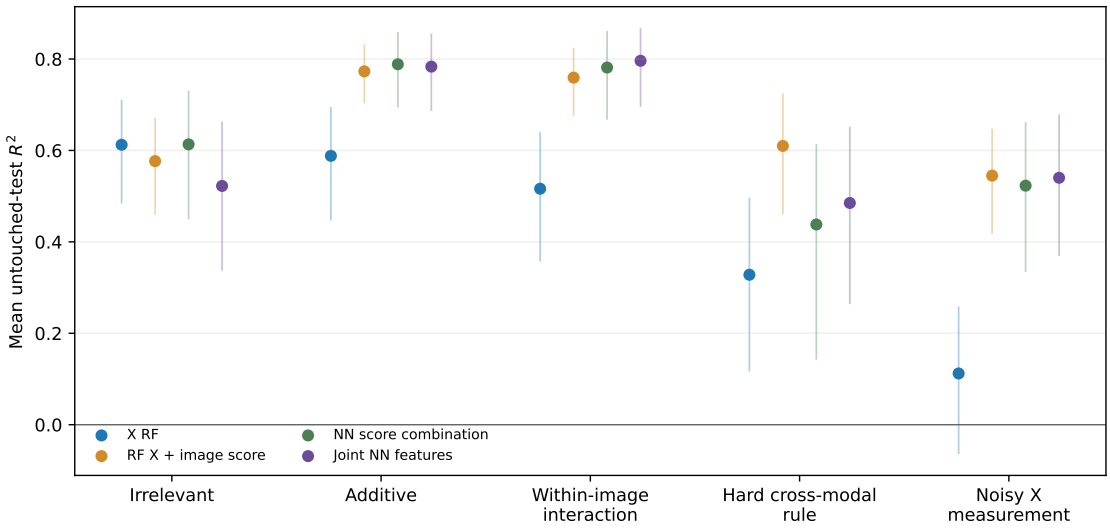


Figure 9: **Random forests in the five Monte Carlo DGPs.** Points report mean untouched-test R^2 over 5,000 replications; whiskers span the 2.5th–97.5th percentiles. The RF score procedure retains the structured variables and adds one leakage-safe scalar image prediction. The same pilot-selected 500-tree configuration is used in every DGP. Existing NN score and joint-NN results use the unchanged simulation protocol.

The forest extension does not simply improve every cell. In Case 1, RF score integration lowers mean R^2 from 0.612 to 0.577. It raises mean R^2 by 0.185 in Case 2, 0.243 in Case 3, 0.282 in Case 4, and 0.433 in Case 5 relative to RF X. The Case-4 level of 0.610 exceeds the joint NN's 0.485. This result materially weakens an interpretation based on necessary information loss from scalar image-score compression. The scalar score retains enough visual-state information for the forest to form the conditional rule with X_2. In Cases 2, 3, and 5, however, the joint linear model remains stronger than RF score integration. No one integration rule dominates across the DGP family.

The sample-size exercise repeats Cases 2–4 at $N \in \{300, 600, 1200, 2400\}$ with 500 replications per cell. It retains the common tuning menu and focuses on score combination, joint linear and neural models, and the oracle NN. Figure 10 shows whether joint flexibility becomes more useful as its estimation cost declines.

The earlier continuous-redundancy exercise answers a separate question and is retained unchanged. It uses one visual shape latent with $\beta_V = 1$ and varies $\mathrm{Corr}(X, V)$ from 0 to .99 at 12 prespecified points. Each grid point uses 1,000 replications and its cached DINO representations. Figure 11 compares the learned increment with the oracle increment. That exercise asks how value declines as useful visual information becomes redundant with X; the four-DGP experiment holds overlap at 0.5 and changes integration architecture.

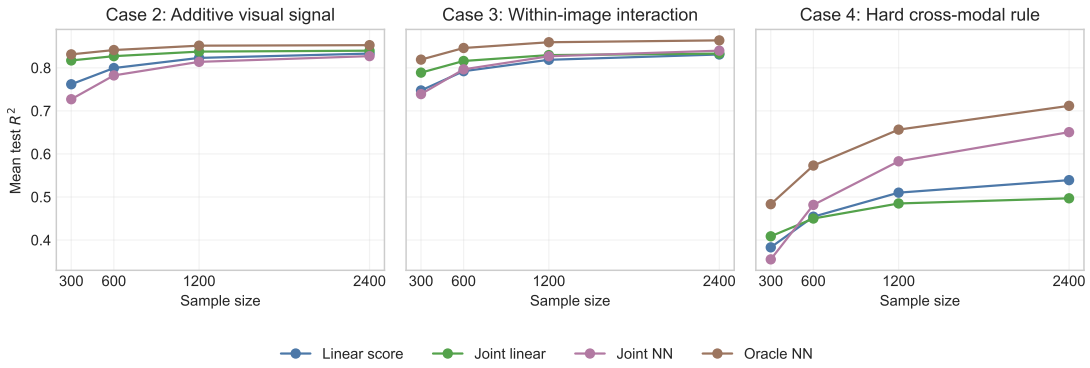


Figure 10: **Integration performance by sample size.** Points report mean test R^2 over 500 replications at each sample size. The DGP, image grid, estimator menu, and validation-only selection rule do not change with N. Lines connect sample sizes within a procedure; they do not order the procedures.

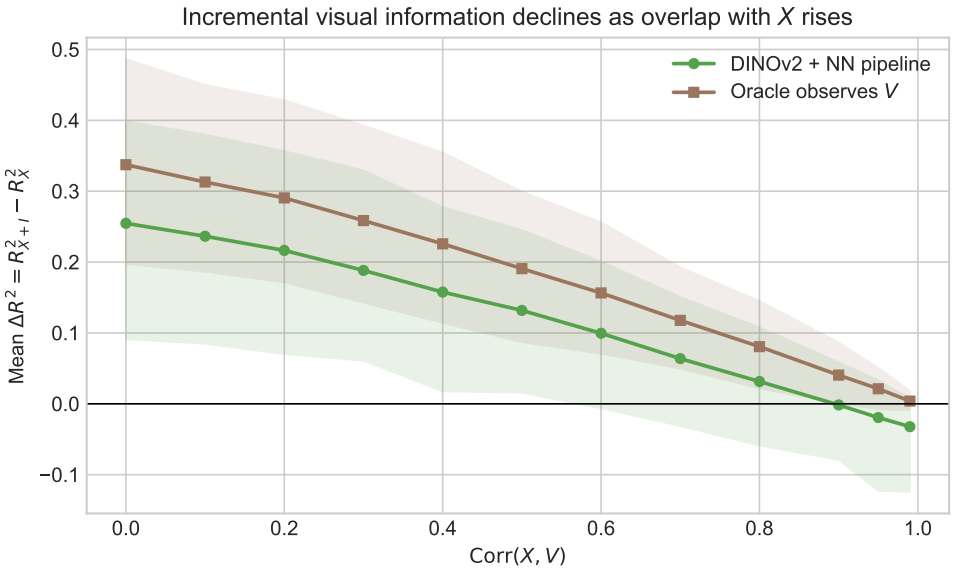


Figure 11: **Incremental value as image information becomes redundant with X.** Points report mean $R^2_{X+I} - R^2_X$ over 1,000 replications at each correlation. The learned pipeline uses cached DINOv2 representations and the fixed NN head; the oracle observes V. Shaded bands span the 2.5th to 97.5th percentiles. The figure describes how residual visual variation declines as X and V become more correlated; it does not impose monotonicity on the estimated pipeline.

A.8 Predictive-overlap regressions

For the preferred image scores, conventional X explains 0.595 in houses and 0.081 in horses on untouched locked-test observations. In salaries, predictive overlap averages 0.111 across the 12 repeated partitions. The house and horse mappings select ridge penalties within validation observations, refit on validation, and predict test image scores. Ontario image scores are group-cross-fitted.

Encoder-specific results reinforce the interpretation. House X predicts ResNet, DINOv2, and SigLIP 2 scores with R^2 0.428, 0.620, and 0.591. Horse X predicts the same scores with 0.040, 0.060, and 0.051. Ontario X predicts them with 0.111, 0.200, and 0.326. These quantities depend on the encoder and downstream head: they measure overlap in a fitted prediction, not an immutable property of the raw image.

A.9 Image prediction of individual covariates

The covariate exercise uses the DINOv2 representation for houses and horses and major Ontario representations for citation-history length. Continuous outcomes use validation-tuned ridge; categorical outcomes use class-weighted logistic regression and validation selection. The same original test split is retained. Building description, garage, and neighbourhood are derived from the actual mutually exclusive indicator blocks. The high-cardinality neighbourhood result should be interpreted cautiously because classes are sparse and some test categories do not appear in every resample.

The horse sale-year result is a useful warning. A side-view horse photograph can reveal camera, backdrop, or digitization changes. The image score is therefore not a direct physical measurement, even when it predicts physical performance. Likewise, house coordinates and neighbourhood are partly predictable from architectural and streetscape regularities. Image-derived covariates can encode institutional and collection processes along with economically meaningful attributes.

A.10 Secondary horse-price results

The sale-price sample includes successfully sold horses with positive observed prices; RNA, withdrawal, and missing-price records are not assigned zero. Log price is the outcome. The structured model includes catalogue variables and actual workout speed where appropriate. The price regressions remain secondary because the paper's horse outcome is physical

Table 14: Can image representations recover individual structured covariates?

Application	Encoder	Covariate	Type	R^2	MAE	Balanced accuracy	Accuracy
Houses	DINOv2	Bedrooms	continuous	0.353	0.540		
Houses	DINOv2	Rooms	continuous	0.062	0.626		
Houses	DINOv2	Washrooms	continuous	0.535	0.724		
Houses	DINOv2	Lot frontage	continuous	0.534	7.303		
Houses	DINOv2	Longitude	continuous	0.096	0.073		
Houses	DINOv2	Latitude	continuous	0.242	0.030		
Houses	DINOv2	Building description	categorical			0.362	0.483
Houses	DINOv2	Garage	categorical			0.384	0.450
Houses	DINOv2	Neighbourhood	categorical			0.088	0.109
Horses	DINOv2	Sale year	continuous	0.389	1.033		
Horses	DINOv2	Hip number	continuous	0.037	261.665		
Horses	DINOv2	Workout distance	continuous	0.104	0.251		
Horses	DINOv2	Sex	categorical			0.526	0.788
Horses	DINOv2	Coat color	categorical			0.871	0.840
Ontario scholars	ResNet50	Citation-history length	continuous	0.274	6.496		
Ontario scholars	DINOv2	Citation-history length	continuous	0.418	5.616		
Ontario scholars	SigLIP 2	Citation-history length	continuous	0.545	4.949		

Notes: Continuous outcomes report test R^2 and MAE; categorical outcomes report balanced accuracy and accuracy. Every model uses the original locked split and validation-only tuning.

performance and because market valuation introduces additional selection and equilibrium interpretation.

Table 15: Secondary evidence from auction prices

Result	Estimate
Image-only log-price best single R2	0.142
Image-only log-price stack R2	0.162
Image-speed score coefficient in price regression	0.019
Image-speed score coefficient in speed regression controlling price	0.243
Held-out speed R2 increment beyond price/catalog	0.049

Notes: Generated-score coefficients are descriptive point estimates; conventional second-stage confidence intervals are not reported. Predictive comparisons use locked held-out observations. These results do not support a causal interpretation.

The image speed score does not materially predict price after actual speed and controls, but it predicts actual speed beyond price and controls. This asymmetry is consistent with the photograph containing physical information not fully capitalized or summarized in transaction price. It does not identify market inefficiency because price reflects expectations, pedigree, risk, bidder composition, and sale conditions that are outside the speed equation.

A.11 Empirical integration-architecture extension

The integration extension is fixed in a machine-readable protocol before the new test comparisons are computed. It retains every original observation, group, split, outcome, structured variable, and frozen embedding. Houses use five modern representation blocks (ResNet50, DINOv2, SigLIP 2, ConvNeXt-B, and ViT-B/16); horses use DINOv2 to isolate integration architecture; Ontario salaries use the three preselected ResNet50, DINOv2, and SigLIP 2 blocks. Train-only principal components reduce each image block, with 16 or 32 components selected by validation MSE. Joint linear models use validation-selected ridge penalties. Joint NNs and structured NNs select among the same three modest architectures and average five prespecified seeds. Score NNs use a separate tiny architecture grid and a group-aware inner split of the original validation observations. No image is re-encoded.

The linear score model combines the previously locked honest validation predictions. The NN score model sees the same two scalar predictions. The joint models instead retain X and reduced image features. A separate compression diagnostic feeds the full X vector and a scalar, five-fold group-cross-fitted image prediction to an NN. This diagnostic prevents an observation's own outcome from constructing its image score, but the resulting coefficient remains a generated-prediction object rather than a structural parameter.

The contrast is application-specific. Houses often reward joint modeling. Horses reward score-level combination; the weaker joint results are consistent with useful DINO information being lost when its 768 coordinates are reduced before combination. Flexible salary models are already strong enough that neither score nor joint feature integration produces a stable increment. A point improvement over structured linear regression can be generated by nonlinear modeling of X rather than by the image.

A.12 Random-forest structured-data robustness

The empirical forest uses the same observations, outcomes, groups, and locked splits as the linear and NN procedures. It receives the same underlying structured information in a tree-appropriate form. House and salary institution indicators remain deterministic. Horse continuous variables use training medians; categorical variables retain group-cross-fitted target encodings and training frequencies. Validation also considers a target-free frequency-only horse variant. Every locked RF prediction averages three prespecified seeds. Validation selects feature subsampling, minimum leaf size, and maximum depth; test outcomes do not enter tuning. For houses, validation selects 67 percent feature subsampling, a minimum leaf

size of one, and unrestricted depth.

For Toronto houses, RF X raises structured test R^2 from 0.876 under the NN to 0.890. The selected joint image model remains stronger at 0.903, leaving an increment of 0.013. For horses, validation retains the structured NN on the locked split, while the forest comparison remains positive in 11 of 12 common repeated selections. Salary repeats select the forest in nine of 12 partitions, and the image null is unchanged. These are comparisons with fitted procedures, not proof that a population-optimal function of X omits the surviving image information.

The RF score procedure is deliberately lower dimensional than the joint image model: it sees X and a group-cross-fitted scalar image prediction. A compact boosted-tree sensitivity is not added to the primary menu. In Toronto, validation favours boosting for both X and $X+$ image score; their locked test R^2 values are 0.892 and 0.903, leaving an increment of 0.011 with interval $[0.000, 0.021]$. Boosting therefore narrows the comparison slightly further. Horse and salary checks do not change their substantive conclusions.

A.13 Repeated partitions and validation selection

The repeated analysis uses the same 12 seed values and the same group-respecting 70/15/15 logic in every application. Within each partition, validation MSE selects X from linear, NN, and random-forest procedures and $X + I$ from linear-score, NN-score, RF-score, joint-linear, and joint-NN procedures. The chosen count matches the prespecified horse stability exercise and makes the complete menu feasible for houses and salaries. It was not increased or decreased after viewing results.

The main qualification is magnitude rather than sign. Horse increments are positive in 11 of 12 partitions, but their mean of 0.0238 is much smaller than the original locked increment of 0.0913. House increments average 0.0069 and are positive in nine partitions. Salary increments average -0.0018 and span both signs. The empirical percentiles describe sensitivity to observed train/validation/test partitions; they are not confidence intervals.

A.14 Image-content falsifications and metadata

The metadata block excludes filenames, paths, identifiers, and sample-order fields. It contains width, height, aspect ratio, file size, brightness, contrast, saturation, RGB means, sharpness, and dark/bright pixel shares. The permutation exercise breaks the image-score/outcome match within validation and test partitions. Both checks use the locked samples because they

Table 16: Validation selection frequencies across repeated splits

Application	Procedure	Count	Share
Academic salaries	structured nn	3	0.25
Academic salaries	structured rf	9	0.75
Horses	structured linear	2	0.17
Horses	structured nn	8	0.67
Horses	structured rf	2	0.17
Houses	structured nn	1	0.08
Houses	structured rf	11	0.92
Academic salaries	joint linear	1	0.08
Academic salaries	linear score	5	0.42
Academic salaries	rf score	6	0.50
Horses	linear score	5	0.42
Horses	nn score	7	0.58
Houses	joint nn	8	0.67
Houses	linear score	2	0.17
Houses	nn score	2	0.17

diagnose content and artifacts rather than partition stability.

Table 17: Image-artifact falsifications under the original linear/NN structured menu

Application	Original selected ΔR^2	Metadata-adjusted ΔR^2	Permuted-score mean ΔR^2
Houses	0.0274	0.0183	-0.0003
Horses	0.0913	0.0823	0.0046
Ontario scholars	-0.0058	0.0013	-0.0002

Notes: These falsifications use the validation-selected linear/NN structured benchmark that preceded the RF extension and are not recomputed against RF X. Metadata include dimensions, aspect ratio, file size per pixel, brightness, contrast, saturation, channel means, sharpness, and exposure shares; no filename or identifier enters. Permutation values average 100 group-aware score permutations with the combination refit on validation outcomes. The horse-only segmentation result is reported separately in the text and Figure 5.

For horses, DeepLabV3 supplies an outcome-independent horse-class mask. The retained horse is placed on a common grey background. The masked image-alone model attains $R^2 = 0.1157$, and selected $X + I$ attains 0.1928. Sale-year prediction falls from 0.5999 to 0.4044. Background and collection practice therefore explain vintage information but not the locked speed increment. Group-bootstrap uncertainty is reported for the original fitted comparison; mask and metadata checks are point-estimate falsifications and are not promoted to repeated-partition confidence statements.

A.15 Salary matching, image availability, and alternative outcomes

Official 2024 and 2025 salary files are retained without modification and verified by SHA-256 hashes. The person-level matching table records normalized names, employer compatibility, method, and review disposition. Salary never resolves ambiguous identity. Of 4,744 audited Ontario profiles, 2,320 match in either year and 2,126 in both. Matching rates vary by institution and matched researchers have longer citation histories and higher citation indices, so inference is restricted to the disclosed matched sample.

The Google graduation-cap default appears exactly in 2,634 source Ontario rows. Six perceptual variants and four blank or generic icons are also removed. The machine-readable audit stores path, hash, dimensions, perceptual distance, and reason. No CRC-specific sample row is read, copied, or analyzed.

On the locked split, image-only R^2 is 0.096 for maximum observed 2024–25 salary, 0.125 for 2025 salary, and 0.144 for the two-year mean. Rich-X integration increments are -0.012, -0.015, and -0.013, respectively. Adding the disclosure title/rank category raises structured fit but does not produce a positive image increment. These alternatives support the repeated-split null without establishing literal zero population information.

A.16 Bootstrap interpretation and generated scores

The paired bootstrap resamples the same locked test groups for each fitted procedure and recomputes their R^2 difference. It conditions on the fitted models and locked partition. It does not integrate over retraining or repartitioning; the repeated-split results serve that separate diagnostic purpose.

Cross-fitting prevents an observation's outcome from training its own image score. It does not eliminate measurement error or make a generated-score coefficient structural (Pagan, 1984; Murphy and Topel, 1985; Christensen and Hansen, 2026; Battaglia et al., 2024). Work on generated labels addresses related second-stage inference problems (Christensen et al., 2026). Our primary object is instead held-out prediction by complete fitted procedures. Conventional second-stage intervals are therefore not attached to descriptive image-score coefficients.

A.17 Noisy structured measurement and predictive-overlap calibration

DGP 5 sets $Y = S + C + \epsilon$ and lets X_1 and X_2 be noisy measurements of the visual latents. At reliability $\rho = 0.5$, mean test R^2 is 0.145 for validation-selected X, 0.602 for image prediction, 0.604 for joint linear integration, and 0.545 for RF-score integration. The image gain declines as ρ increases from 0.3 to 0.9. Images can therefore add value when X records the same concept noisily; positive predictive value need not imply that the concept is absent from structured data.

Table 18: Noisy-measurement DGP: structured reliability and image value

ρ	R^2_X	R^2_I	R^2_{X+I}	ΔR^2_I
0.30	0.033	0.595	0.584	0.551
0.50	0.143	0.604	0.596	0.453
0.70	0.301	0.596	0.587	0.286
0.90	0.525	0.601	0.598	0.073

Notes: Each row reports means from 500 replications of DGP 5 at the stated correlation between each structured measurement and its visual latent. The image grid, DINOv2 representation, estimator menu, and validation-only selection rule are fixed across rows. $X + I$ is the validation-selected integration procedure. The main 5,000-replication design uses $\rho = 0.5$ and gives $R^2_X = 0.145$, $R^2_I = 0.602$, and $R^2_{X+I} = 0.593$.

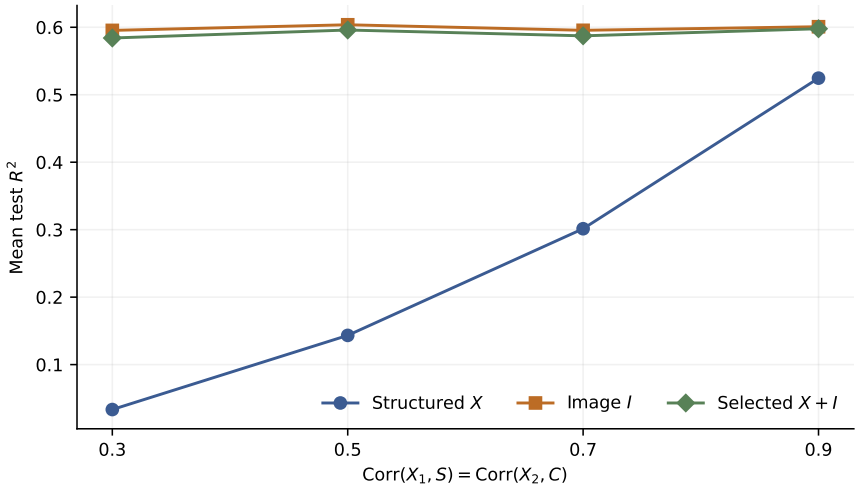


Figure 12: **Image value as structured measurement improves.** Points report mean untouched-test R^2 over 500 replications at each reliability. The photograph and X measure the same underlying traits; only structured measurement reliability changes.

The continuous-redundancy experiment supplies a second calibration. Fitted predictive

overlap rises as the true overlap increases, while incremental image value declines. At high overlap the learned increment can become negative in finite samples even though optimal population risk cannot worsen when information is added.

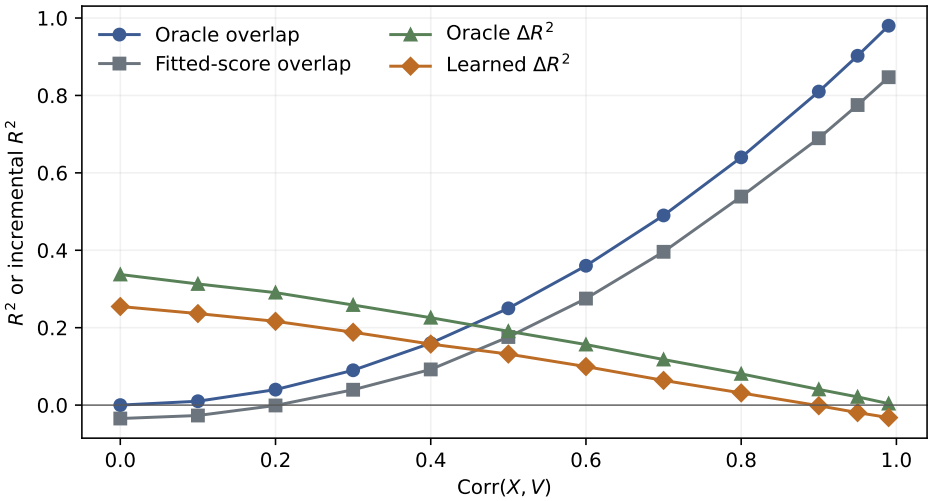


Figure 13: **Predictive overlap and incremental image value as redundancy varies.** Oracle overlap equals ρ^2. Fitted overlap predicts the fitted image score from X. Learned and oracle increments use the same cached DINOv2 representations across the redundancy grid.

A.18 Computational cost

Table 19: Frozen-encoder computational costs

Encoder	Parameters	Dimension	Images/sec	Cost vs ResNet	Peak GPU GB	Cache MB
ResNet50	23,587,712	2048	237.914	1.000	0.647	59.581
DINOv2-B/14	86,580,480	768	148.197	1.605	0.838	22.343
SigLIP 2-B/16	92,884,224	768	162.615	1.463	1.730	22.343
ConvNeXt-B	87,566,464	1024	128.437	1.852	0.946	29.790
ViT-B/16	85,798,656	768	128.108	1.857	0.689	22.343
VGG16-GAP	14,714,688	512	56.176	4.235	2.407	14.895
InceptionV3-GAP	21,785,568	2048	62.350	3.816	0.706	59.581
MobileNet (legacy)	3,228,864	1024				29.790
COCO panoptic		218				6.342
ADE20K panoptic		232				6.749

Notes: Timing uses the same machine where available. Historical semantic representations lack comparable extraction timings. Downstream NN-head training is small relative to encoder extraction because embeddings are cached.

The cost table separates representation extraction from downstream fitting. ResNet50 processes about 238 house images per second in the same-machine benchmark. DINOv2 and

SigLIP 2 are roughly 1.6 and 1.5 times the ResNet inference cost but are much stronger for the house and horse applications. Full multi-encoder unions cost several encoders per observation. Once embeddings are cached, ridge and small NN heads are inexpensive. This distinction supports caching and validation before deployment: an applied researcher can compare heads cheaply, but an $X + I$ system must still pay the encoder cost for every new image.

A.19 Inference and generated-score caveats

The paired bootstrap resamples the same groups for each competing model and recomputes R^2 and its difference. It treats the fitted prediction functions and split as fixed. It therefore addresses whether the observed locked test set supports a gain, not how performance would vary over every retraining and repartitioning. For NN heads, seed-averaged predictions reduce optimization noise, and the separate representation-risk ablation reports seed dispersion.

Cross-fitting prevents an observation's outcome from directly training its own image score. It does not make generated covariates observed without error. Classical two-step and generated-regressor concerns remain (Pagan, 1984; Murphy and Topel, 1985). Recent work develops corrections for ML-generated covariates (Fong and Tyler, 2021; Christensen and Hansen, 2026). Because our primary measure is a predictive loss difference, we avoid treating the score coefficient as structural. Any reported generated-score coefficients describe associations; conventional second-stage confidence intervals are not reported for them.

A.20 Replication and data access

Replication materials will be released separately, subject to applicable data-access restrictions. The Ontario copy includes only artifacts whose use remains subject to the terms of the original Google Scholar collection; it excludes the CRC sample entirely.

The Toronto MLS records and listing photographs and the OBS catalogue records and horse photographs are not redistributed by this paper. They remain subject to their providers' access and use restrictions.

A.21 Leakage and quality-control checklist

The final verification applies the following checks:

1. Every headline observation ID appears exactly once in its application manifest.
2. No property address, horse group, researcher group, or exact image hash crosses train, validation, and test.
3. All saved embeddings are finite and match the manifest row digest and declared dimension.
4. Structured scaling, imputation, PCA, and frequency/target encodings use training data only.
5. NN early stopping, architecture selection, image-model selection, and score-combination weights use validation outcomes only.
6. Test outcomes are used only for final metrics and paired resampling.
7. Cross-fitted image scores exclude their own outcome from head training.
8. Ontario source copies contain only `source_group == ontario`; the CRC row count in the new project is zero.
9. Every Ontario placeholder removal has a machine-readable reason, and removed/retained contact sheets were visually checked.
10. The machine-readable central table, manuscript text, and LaTeX tables agree to displayed precision.